

# Low-Dimensional Topology Reading Group

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These are a collection of notes used for a low-dimensional topology reading group. The target audience is undergrads with some proof experience and little exposure to topology. Whenever possible, I will choose lack of rigor over assumed background in point-set topology, algebra, analysis, etc. However, I will make frequent use of set-theoretic notation — if you are unfamiliar with this, please let me know, and I can get you up to speed on everything you need to know. Exercises are sprinkled throughout the exposition, and they are designed to get the reader thinking about the contents and visualizing the objects of study. Exercises that are challenging or more for fun will be denoted by an asterisk (\*).

If you have any trouble understanding anything, get stuck on an exercise, or have any questions, please reach out to me! I have a place in the Discord for people to ask questions, and if you'd prefer, my DMs are always open. Also, please let me know if you find typos or mistakes (which there are guaranteed to be).

I am doing the sections on Riemann surfaces and algebraic topology last, since I am not sure if we'll have time for those.

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# 1 Intro to Manifolds

This section gives a brief, “formal” introduction to manifolds. I will roughly follow the exposition given in Guillemin and Pollack’s book on *Differential Topology*. The only background I am assuming is calculus and some discrete math, though things will probably be more intuitive if you have some experience with real analysis (specifically metric space topology).

## 1.1 Motivation

Before we begin, a natural question to ask is, why should we care about manifolds? Why are they something worth studying at all? For those more interested in applications, one popular use of these objects is in physics. The language of physics is calculus, and calculus is built on the local geometry of Euclidean space. Thus, this language most naturally adapts itself to spaces that locally “look like” Euclidean space; we call these objects *manifolds*. The Hamiltonian formulation of classical mechanics occurs within what’s called a *symplectic* manifold. According to general relativity, we live on what can be seen as a 4-dimensional manifold, with three spatial dimensions and one temporal dimension. With this, the geometry of such an object becomes immediately relevant to the real world. A disclaimer however, we will not be focusing on the *geometry* of these spaces, we will be focusing on their *topology*.

Fundamentally, geometry is the study of properties of spaces such as distance, shape, volume, relative position, and so on. Think of Euclidean geometry, where we might study triangles in the plane up to congruence. If we generalize this notion to allow for scaling, we are studying triangles up to similarity. In order to start doing topology, we generalize even further, allowing for *any* continuous deformation of our triangle. The only thing we exclude are closing or opening holes, tearing, gluing, or self-intersection. We find that triangles become very boring, since up to this notion of equivalence,<sup>1</sup> all triangles are the same. So, what does the topology of a space tell us? There are plethora of important properties preserved under

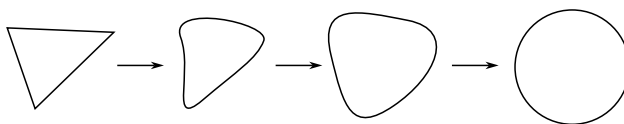


Figure 1.1: Continuously deforming a triangle into a circle

the operations we’ve described: dimension, number of “holes”, connectedness, compactness, etc. Although a triangle is topologically the same as a circle, it is topologically different than a closed interval.

There is a deep interplay between the geometry of manifolds and their topology. The topology of a manifold can determine what kinds of geometry can be put on them (flat, spherical, hyperbolic, etc.), and specific properties of that geometry, for instance via the Gauss-Bonnet theorem. The study of this interplay is called *geometric topology*. Hopefully, I’ll be able to tell you a bit about these connections throughout these notes.

So physicists (and some engineers) care about topology, specifically the topology of manifolds. Why should a mathematician care? Topology can yield beautiful, deep insights into the properties of many geometric objects that mathematicians have studied for thousands of years. Some ideas from the subject date back all the way to Euler, where he proved that it was impossible to find a route through the town of Königsberg that would cross each of its seven bridges exactly once. This result did not depend on the lengths of the bridges or the distances from one another, but only on the property of which bridges were connected to which islands or riverbanks. Manifolds are useful objects of study in almost every branch of math dealing with geometry or topology.

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<sup>1</sup>Called *homeomorphism*.

## 1.2 Basic Topology

Before we come to understand manifolds, we need a crashcourse in topology. In particular, we will be focusing on the topology of subsets of Euclidean space  $\mathbb{R}^n$  (here,  $n$  and  $m$  will be used freely as some natural numbers referring to the dimensions of these spaces). This is primarily to avoid the unnecessary abstraction one encounters in a class on point-set topology, and it's really all we need to get our hands dirty with manifolds. We need to introduce the terminology of *open* and *closed* sets — we do so in terms of *balls*.

**Definition 1.1.** Let  $p$  be a point in the Euclidean space  $\mathbb{R}^n$ . The **ball of radius  $r$  around  $p$**  is the set

$$B_r(p) = \{q \in \mathbb{R}^n : d(p, q) < r\},$$

where  $d(p, q)$  is the usual Euclidean distance between  $p$  and  $q$ . Some may refer to this as the **open ball**, and correspondingly we can define the **closed ball**

$$\overline{B}_r(p) = \{q \in \mathbb{R}^n : d(p, q) \leq r\}.$$

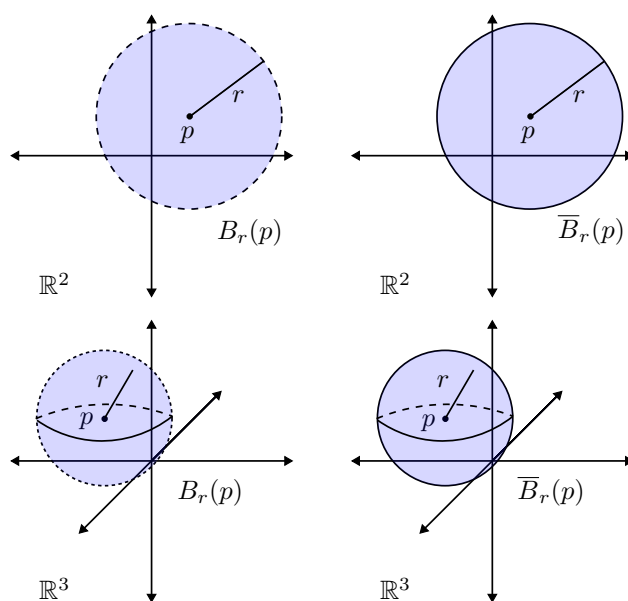


Figure 1.2: Open and closed balls in  $\mathbb{R}^2$  and  $\mathbb{R}^3$

**Definition 1.2.** A subset  $X$  of  $\mathbb{R}^n$  is **open** if for each point  $p$  of  $X$ , there exists some radius  $r > 0$  such that  $B_r(p)$  is contained entirely within  $X$ . We say  $X$  is **closed** if its complement  $\mathbb{R}^n \setminus X$  is open. A **neighborhood** around a point  $p \in \mathbb{R}^n$  is an open set  $U$  such that  $p \in U$ .

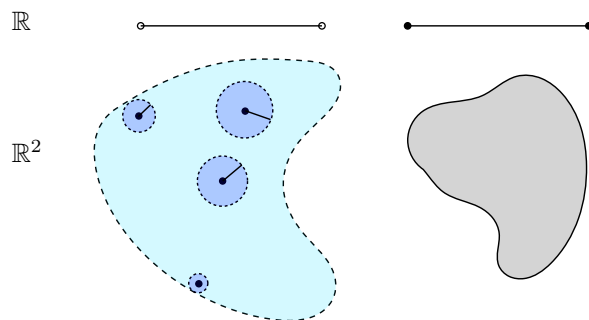


Figure 1.3: Open and closed sets in  $\mathbb{R}$  and  $\mathbb{R}^2$

**Exercise 1.1.** Here are some good problems to practice with these definitions if they're new to you:

- (1) Prove that any open ball  $B$  is open.
- (2) Prove that any closed ball  $\overline{B}$  is closed. This will involve you choosing a point in the complement  $\mathbb{R}^n \setminus \overline{B}$  and finding an open ball around it which is disjoint from  $\overline{B}$ .
- (3) Prove that for open sets  $U, V \subset \mathbb{R}^n$ , we have that  $U \cup V$  and  $U \cap V$  are open.
- \* (4) Prove that for open sets  $U \subset \mathbb{R}^n$  and  $V \subset \mathbb{R}^m$ , we have that

$$U \times V = \{(u, v) : u \in U, v \in V\}$$

is open in  $\mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$

- (4) Can a subset  $X \subset \mathbb{R}^n$  be open and closed at the same time? Can it be neither? If so, find some examples.

We can also define notions of open and closed which are relative to a subset of Euclidean space; this is particularly important when we get to the definition of manifolds.

**Definition 1.3.** If  $U \subset X \subset \mathbb{R}^n$ , then we say that  $U$  is **open in  $X$**  if it can be written as  $U = V \cap X$  for some open subset  $V$  of  $\mathbb{R}^n$ . Relative closedness is defined similarly.

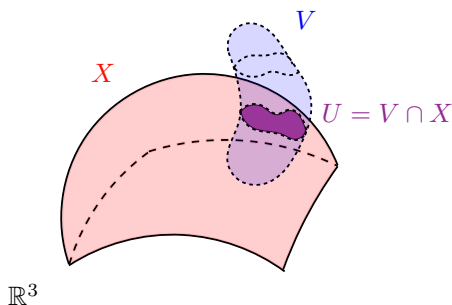


Figure 1.4: A subset  $U$  which is relatively open in  $X \subset \mathbb{R}^3$

**Exercise 1.2.** Show that  $X$  is open and closed in itself.

It is natural to wonder why these definitions are important, since I haven't done much to motivate them so far. For our purposes, the main reason we care about openness and closedness is because open subsets of  $\mathbb{R}^n$  are the only place it really makes sense to do calculus:

**Definition 1.4.** Let  $U \subset \mathbb{R}^n$  be open. A function  $f : U \rightarrow \mathbb{R}^m$  is **smooth** if it has continuous partial derivatives of all orders.

If the domain of  $f$  were not open, it would not always make sense to ask about partial derivatives (why?). So, we need to be clever if we want to define smooth maps on arbitrary subsets of  $\mathbb{R}^n$ . It might be worth taking some time to dissect the following definition, and to understand why it's the right one for the job.

**Definition 1.5.** For any subset  $X \subset \mathbb{R}^n$ , a function  $f : X \rightarrow \mathbb{R}^m$  is **smooth** if it restricts to a smooth map on a neighborhood of each point in  $X$ . That is,  $f$  is smooth if for all  $p \in X$ , there is a neighborhood  $U$  of  $p$  and a smooth map  $F : U \rightarrow \mathbb{R}^m$  such that  $F$  and  $f$  are the same on  $U \cap X$ .

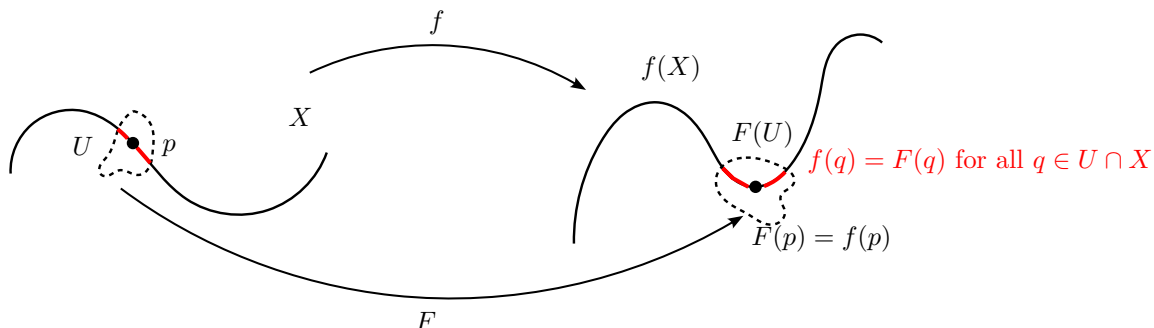


Figure 1.5: A smooth map between paths in  $\mathbb{R}^2$

**Definition 1.6.** A smooth map  $f : X \rightarrow Y$  where  $X \subset \mathbb{R}^n$  and  $Y \subset \mathbb{R}^m$  is a **diffeomorphism** if it is a bijection and if the inverse map  $f^{-1} : Y \rightarrow X$  is also smooth. If a diffeomorphism exists, we say  $X$  and  $Y$  are **diffeomorphic**; sometimes we will write  $X \cong Y$  or  $X \xrightarrow{\sim} Y$ .

Philosophically, diffeomorphic subsets are “the same” topologically, even though they might be different as sets or as geometric objects. With some practice, you’ll be able to easily identify when two subsets of Euclidean space are diffeomorphic. Here are some examples from Guillemin and Pollack to help build intuition:

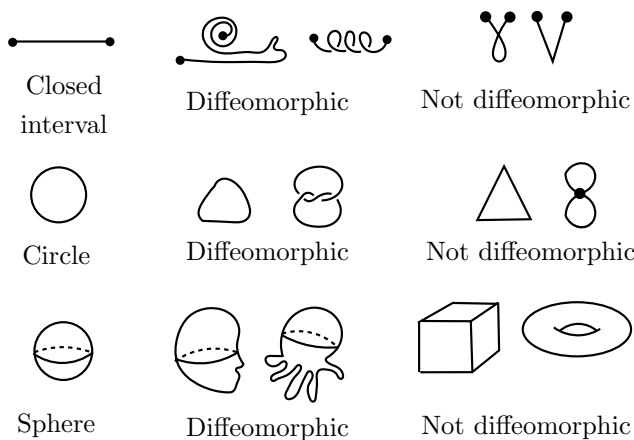


Figure 1.6: Examples of diffeomorphic subsets

**Exercise 1.3.** Make sure you understand why each of the above examples are or are not diffeomorphic to one another. Where are the problem points that break our desired smoothness property? Why are we able to twist our subsets into knots via a diffeomorphism — wouldn’t that require some kind of discontinuity?

**\*Exercise 1.4.** You may have noticed that in our definition of diffeomorphism, we had to explicitly require that  $f^{-1}$  also be smooth. Can you think of any examples of bijective smooth maps whose inverse is not smooth?

**Exercise 1.5.** Suppose that  $X$  is a subset of  $\mathbb{R}^N$  and  $Z$  is a subset of  $X$ . Show that any smooth map on  $X$  restricts to a smooth map on  $Z$ . Show that a diffeomorphism of  $X$  restricts to a diffeomorphism of  $Z$ .

**Exercise 1.6.** Let  $X \subset \mathbb{R}^N$ ,  $Y \subset \mathbb{R}^M$ ,  $Z \subset \mathbb{R}^L$  be arbitrary subsets, and let  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  be smooth maps. Prove that the composition  $g \circ f : X \rightarrow Z$  is smooth. If  $f$  and  $g$  are both diffeomorphisms, prove  $g \circ f$  is as well.

### 1.3 Manifolds

Manifolds are particularly nice spaces. The most familiar examples are the ones you would see in an introductory topology class or pop math YouTube video: surfaces like the sphere or torus (the crust of a doughnut). Each point on these surfaces is contained in a small, curved disk that can be smoothly flattened into a disk in the plane. So, if we zoom in far enough, each point basically looks the same; they look like they're contained in a small section of  $\mathbb{R}^2$ .

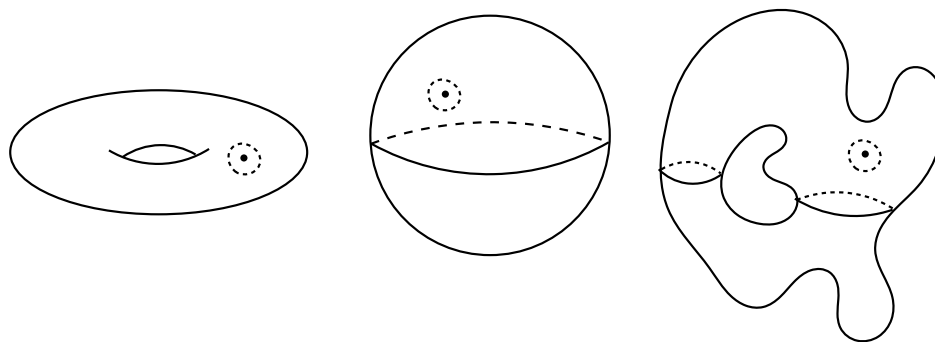


Figure 1.7: Examples of surfaces

Although we need 3 dimensions to see these shapes,<sup>2</sup> they're locally<sup>3</sup> the same as the plane or some other infinite surface, so we say they are 2-dimensional manifolds — *surfaces* for short. One subtle nonexample is the infinite cone.

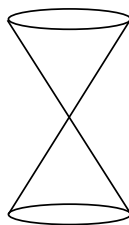


Figure 1.8: A subtle nonexample

Around most points we have our nice small disks, but no neighborhood of the cone point looks like a simple piece of the plane.

To make this intuitive idea formal, we use the above notions of smoothness.

<sup>2</sup>Some people call this the *embedding dimension*. This is the smallest dimension such that we can see the manifold in its entirety.

<sup>3</sup>In basically all contexts, the word “locally” refers to a property that is true in some neighborhood of every point in a space.

**Definition 1.7.** Suppose that  $X$  is a subset of some big, ambient Euclidean space  $\mathbb{R}^N$ . Then  $X$  is a  **$k$ -dimensional manifold** — or a  **$k$ -manifold** for short — if it is **locally diffeomorphic** to  $\mathbb{R}^k$ , meaning that for each point  $p \in X$ , there is an open set  $U$  in  $X$  containing  $p$  which is diffeomorphic to an open subset  $V$  of  $\mathbb{R}^k$ . These are our “small disks” from above. We call a diffeomorphism  $\varphi : U \rightarrow V$  a **coordinate system** on  $U$ . Sometimes, we might refer to  $U$  itself as a **coordinate chart** or **coordinate patch**. When we write  $\varphi$  as its component functions  $\varphi = (x_1, \dots, x_k)$ , the  $k$  smooth functions  $x_1, \dots, x_k$  on  $U$  are called **coordinate functions**. The inverse diffeomorphism  $\varphi^{-1} : V \rightarrow U$  mapping a piece of Euclidean space onto our manifold is called a **parametrization** of the neighborhood  $U$ .

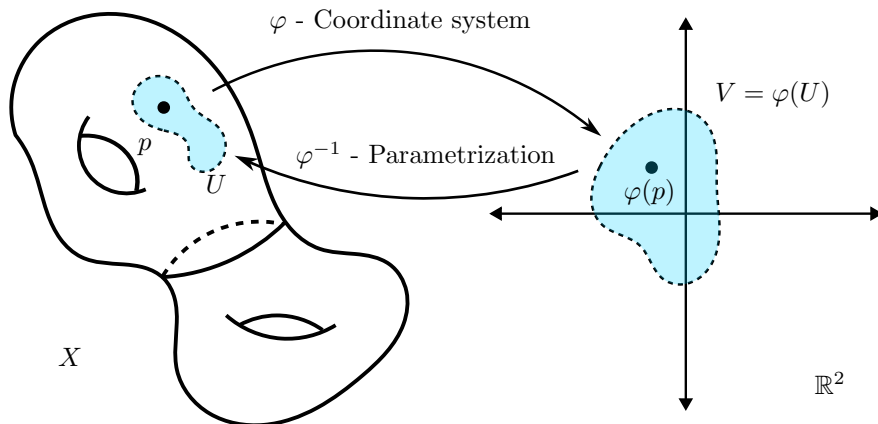


Figure 1.9: Around each point  $p$  in  $X$ , there should be a neighborhood  $U$  of  $p$  and a coordinate system on  $U$

Let’s work through an example to see how this definition works in practice. I will trace out the first steps in detail before being a bit more hand wavy in the later bits.

**Proposition 1.1.** The circle

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

is a 1-dimensional manifold.

*Proof.* Consider the upper semicircle  $N = \{(x, y) : y > 0\}$ , which is open in the circle. First, we see that the projection map  $\varphi : N \rightarrow \mathbb{R}$  given by  $\varphi(x, y) = x$  is a diffeomorphism onto the open interval  $(-1, 1)$ . Indeed, given any  $-1 < t < 1$ , there is a point  $(t, \sqrt{1-t^2}) \in S^1$  mapping to it, so  $\varphi$  is onto. Moreover, there is only one such point solving the equation  $t^2 + y^2 = 1$  where  $y > 0$ , hence  $\varphi$  is one-to-one. The function  $\varphi$  is smooth because it is simply a linear function, and  $\varphi^{-1}(t) = (t, \sqrt{1-t^2})$  can be seen to be smooth on its domain as well. Therefore,  $\varphi$  is a smooth bijection with a smooth inverse, thus a diffeomorphism. Consequently, in the case of a point  $(x, y) \in N$ , we have an open set in  $S^1$  containing  $(x, y)$  and a diffeomorphism of this open set onto  $(-1, 1) \subset \mathbb{R}$ .

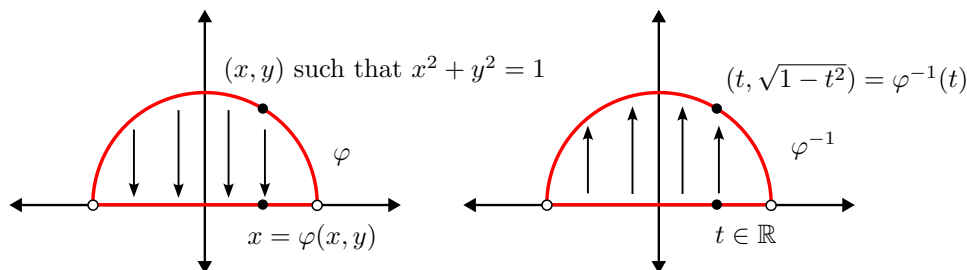


Figure 1.10: A coordinate chart around each point in  $N$



By some abuse of notation, the same map  $\varphi : S \rightarrow \mathbb{R}$  on the lower semicircle  $S = \{(x, y) : y < 0\}$  does the same for points in  $S$  ( $\varphi$  is actually a smooth map on all of  $\mathbb{R}^2$  onto  $\mathbb{R}$ !). This leaves only the two axis points  $(-1, 0)$  and  $(1, 0)$ . But we can just use projection from the left and right semicircles  $E, W$  onto the  $y$ -axis  $\psi : E \rightarrow \mathbb{R}$  and  $\psi : W \rightarrow \mathbb{R}$  where  $\psi(x, y) = y$ , to get diffeomorphisms from  $E, W$  to  $(-1, 1)$ . So, given any point on the circle, we have a local diffeomorphism onto  $\mathbb{R}$ , and we find that  $S^1$  is a 1-dimensional manifold.  $\square$

**Exercise 1.7.** In the proof, we used 4 coordinate charts to cover every point on  $S^1$ . Can we do it with less? Prove that the whole circle cannot be parametrized by a single map.

**\*Exercise 1.8.** Generalize the argument in the proof to show that the  $n$ -sphere

$$S^n = \left\{ (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : \sqrt{x_1^2 + \dots + x_{n+1}^2} = 1 \right\}$$

is an  $n$ -dimensional manifold. How many coordinate charts do we need in general?

A useful method of producing new manifolds is the *Cartesian product*. Suppose that  $X$  and  $Y$  are  $k$ - and  $\ell$ -manifolds inside of  $\mathbb{R}^M$  and  $\mathbb{R}^N$ , respectively. Then

$$X \times Y = \{(x, y) : x \in X, y \in Y\}$$

is a subset of  $\mathbb{R}^M \times \mathbb{R}^N = \mathbb{R}^{M+N}$ . Given coordinate maps  $\varphi : U \rightarrow \mathbb{R}^k$  and  $\psi : V \rightarrow \mathbb{R}^\ell$  with  $U \subset X$  and  $V \subset Y$  open in  $X$  and  $Y$ , respectively, we can define a new coordinate system

$$\varphi \times \psi : U \times V \rightarrow \mathbb{R}^k \times \mathbb{R}^\ell \quad \text{given by} \quad (\varphi \times \psi)(u, v) = (\varphi(u), \psi(v)).$$

Of course, as per Exercise 1.1,  $\varphi(U) \times \psi(V)$  is open in  $\mathbb{R}^{M+N}$ . So, we find that

**Theorem 1.1.** If  $X$  and  $Y$  are  $k$ - and  $\ell$ -manifolds, respectively, then  $X \times Y$  is a  $(k + \ell)$ -manifold.

**Exercise 1.9.** Carefully complete the proof for Theorem 1.1 — check specifically that the map  $\varphi \times \psi$  is a coordinate system.

Now, let's see some quick examples:

- (1) As you can see from the figure, there are two ways to read any given Cartesian product: left to right or right to left.

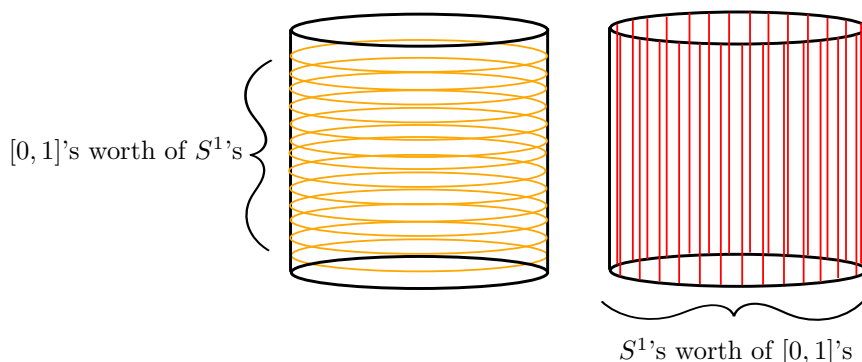


Figure 1.11:  $S^1 \times [0, 1]$

- (2) In the following, we again see a red curve worth of orange curves and an orange curve worth of red curves.

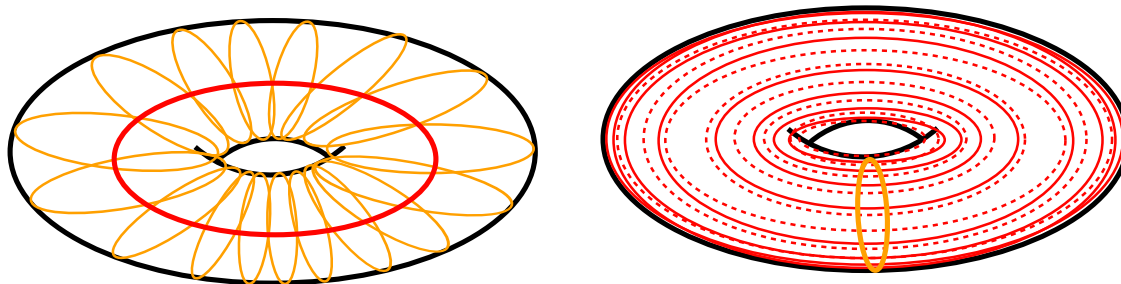


Figure 1.12:  $S^1 \times S^1$

**\*Exercise 1.10.** A torus is the set of points in  $\mathbb{R}^3$  at a distance  $s$  from the circle of radius  $r$  in the  $xy$ -plane, where  $0 < s < r$ . Prove that these tori are all diffeomorphic to  $S^1 \times S^1$ . Draw the cases where  $s = r$  and  $s > r$ . Why are these spaces not manifolds?

**\*Exercise 1.11.** Show that the projection map  $X \times Y \rightarrow X$  given by  $(x, y) \mapsto x$  is smooth.

One last useful notion that we will be using later is that of a *submanifold*. If  $X$  and  $Z$  are both manifolds in  $\mathbb{R}^N$  and  $Z \subset X$ , then  $Z$  is a **submanifold** of  $X$ . In particular,  $X$  itself is a submanifold of  $\mathbb{R}^N$ .

**Exercise 1.12.** Show that any open set  $U$  in a manifold  $X$  is a submanifold of  $X$ .

Here are some fun but optional exercises:

**\*Exercise 1.13.** Stereographic projection is a map  $f$  from the punctured sphere  $S^2 \setminus \{N\}$  onto  $\mathbb{R}^2$ , where  $N$  is the north pole at  $(0, 0, 1)$ . For any  $p \in S^2 \setminus \{N\}$ , the point  $f(p)$  is defined to be the point at which the line through  $N$  and  $p$  intersects the  $xy$ -plane:

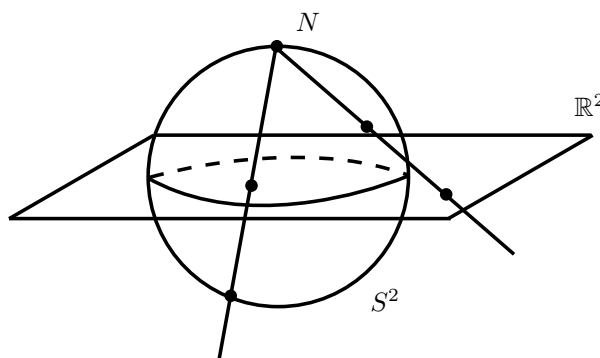


Figure 1.13: Stereographic Projection

One finds that the whole southern and northern hemispheres are mapped to the interior and exterior of the equator, respectively, and the equator is fixed in place. Prove that  $f$  is a diffeomorphism by explicitly specifying the coordinate functions for  $f$  and solving for  $f^{-1}$ . With this, we can essentially think of  $S^2$  as  $\mathbb{R}^2$  with an added “point at infinity”.

**\*Exercise 1.14.** Generalize the above construction to show that  $S^n \setminus \{N\}$  is diffeomorphic to  $\mathbb{R}^n$ . How does this resolve Exercise 1.8?

## 2 More Manifolds

In the previous section, we saw some prominent examples of manifolds:  $n$ -spheres, tori, Cartesian products, etc. In these next few sections, we will explore dimensions 0, 1, and 2. In fact, it is our goal to classify all “reasonable” manifolds in these dimensions, as well as to explain what “reasonable” means in this context.

### 2.1 Properties of Manifolds

Before we begin, it is worth noting: we generally consider manifolds *up to diffeomorphism*, namely, two manifolds are “the same” if there exists a diffeomorphism between them. So, when we talk about classification of manifolds, we really mean classification of diffeomorphism *classes* of manifolds. Whenever we define a “property” of a manifold, it should be invariant under diffeomorphism; if there exists a diffeomorphism between two manifolds, they should both have the same value for that property. I have assumed this implicitly a number of times already:

**Exercise 2.1.** Show that if two manifolds are diffeomorphic, they have the same dimension.

Although we may not have thought about it yet, there is something of a trivial way of constructing new examples of manifolds: disjoint union. Two circles sitting next to each other is a different manifold than one circle (how could we prove this?). A circle and a line next to it is a different manifold than two circles. It does seem a bit silly to be worried about classifying these sorts of manifolds, so we usually restrict to the case where the manifold is *connected*.

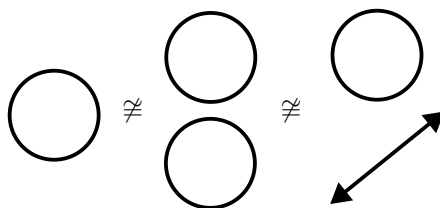


Figure 2.1: None of the pictured manifolds are diffeomorphic

**Definition 2.1.** A subset  $X \subset \mathbb{R}^n$  is **connected** if it cannot be expressed as the union of two non-empty disjoint subsets  $U, V \subset X$  which are open in  $X$ . Be careful here, open *in*  $X$  is an important distinction; see Exercise 1.2.

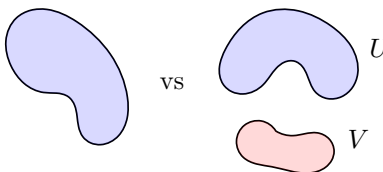


Figure 2.2: Connected vs. disconnected

**Exercise 2.2.** Prove that  $S^1$  is connected (using the definition above). One thing to note here is that, frequently, it is difficult to prove a negative: “cannot be expressed as the union of two disjoint open sets” — it might be useful to try a different proof technique like contradiction or contraposition.

**Exercise 2.3.** Let  $X \subset \mathbb{R}^n$  and  $f : X \rightarrow \mathbb{R}^m$  be a smooth map. Show that if  $X$  is connected, then its image  $f(X)$  is also connected. This also proves that connectedness is a diffeomorphism invariant.

There is actually a stronger and potentially more intuitive definition for connectedness that works well in this context:

**Definition 2.2.** A subset  $X \subset \mathbb{R}^n$  is **path connected** if for any two points  $p, q$  in  $X$ , there exists a smooth function<sup>4</sup>  $\gamma : [0, 1] \rightarrow X$  such that  $\gamma(0) = p$  and  $\gamma(1) = q$ . We call functions in the same form as  $\gamma$  **paths**.

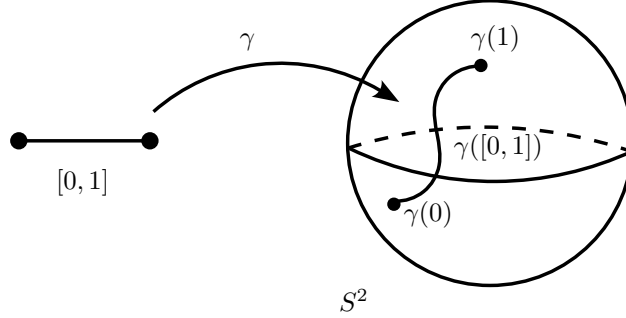


Figure 2.3: The sphere is path connected

This definition is useful, because it is often easier to prove path connectedness than connectedness, and

**Theorem 2.1.** If a subset  $X \subset \mathbb{R}^n$  is path connected, then it is also connected.

*Proof.* Let  $X$  be path connected. Suppose for contradiction that we can write  $X$  as the union of two open sets  $U, V$  in  $X$  such that  $U \cap V = \emptyset$ . Pick any points  $p \in U$  and  $q \in V$ , and let  $\gamma : [0, 1] \rightarrow X$  be a path between them. Consider the subsets  $\gamma^{-1}(U)$  and  $\gamma^{-1}(V)$  — these are nonempty since  $0 \in \gamma^{-1}(U)$  and  $1 \in \gamma^{-1}(V)$ . Since  $U$  and  $V$  cover  $X$ , their preimages cover  $[0, 1]$ . Moreover, since  $\gamma$  is smooth, they are disjoint and open. This contradicts that  $[0, 1]$  is connected, and the theorem holds.<sup>5</sup>  $\square$

Note, the converse does not hold in general.<sup>6</sup> However, in the case of manifolds, it does!

**Theorem 2.2.** A manifold  $X$  is connected if and only if it is path connected.

*Proof.* Thanks to Theorem 2.1, we need only show the forward implication. Let  $X$  be a connected manifold, and let  $p$  be a point in  $X$ . Consider the set

$$U = \{q \in X : \text{there is a path from } p \text{ to } q\}.$$

We call this type of subset a **path component** of  $X$ . I claim that  $U$  and its complement  $\bar{U} = X \setminus U$  are *both* open in  $X$ . If the claim were true, then it must be that  $X = U$  and  $\bar{U} = \emptyset$ , since  $X$  is the union of  $U$  and  $\bar{U}$ ,  $U$  and  $\bar{U}$  are disjoint, and  $U$  is nonempty.<sup>7</sup>

Now, we prove the claim. Let  $q \in U$ . Since  $X$  is a manifold, there is a neighborhood  $V$  of  $q$  which is diffeomorphic to an open ball in  $\mathbb{R}^n$ ; in particular,  $V$  is path connected. Note, it must be that  $V \subset U$ , since if not, we would be able to draw a path from some point in  $\bar{U} \cap V \neq \emptyset$  to  $q$ , a contradiction. This proves that  $U$  is open, since we can find an open neighborhood around each point in  $U$ . The argument for  $\bar{U}$  is similar: for  $q \in \bar{U}$ , we find a path connected neighborhood  $V \subset \bar{U}$  because if not, we could draw a path from a point in  $U \cap V \neq \emptyset$  to  $q$ . This proves the claim.  $\square$

<sup>4</sup>In the actual definition, we only require  $\gamma$  to be *continuous*, but smooth works fine for our purposes.

<sup>5</sup>The technique we used in this proof of “pulling back” some structure on the target space of some map is ubiquitous in topology.

<sup>6</sup>A famous counterexample is the *topologist’s sine curve*. Look this up if you’re interested!

<sup>7</sup>This is also a very common proof technique in topology — proving that some subset is open and closed and concluding something about connectedness.

Now, we should spend some time discussing *compactness*. This property of manifolds is very important, and we'll see why soon. There is a much more general (and more confusing) way of stating this definition, so I hope the one I give here captures the intuition fairly well.

**Definition 2.3.** A subset  $X \subset \mathbb{R}^n$  is **compact** if it is *closed* (as a subset of  $\mathbb{R}^n$ ) and *bounded*. There are many equivalent ways of defining *boundedness*; here are two of them:

- $X$  is **bounded** if there exists some  $r > 0$  such that for all  $p, q$  in  $X$ , we have that  $d(p, q) < r$ , where  $d$  again denotes the usual Euclidean distance.
- $X$  is **bounded** if for some point  $p$  in  $X$ , there exists  $r > 0$  such that  $B_r(p)$  contains all of  $X$ .

**\*Exercise 2.4.** Prove that the two definitions of bounded are the same.

The proof that compactness is a diffeomorphism invariant necessarily uses tools we have not developed from point-set topology, so I will omit it. Let's see some examples of compact and noncompact manifolds to get a feel for the definition:

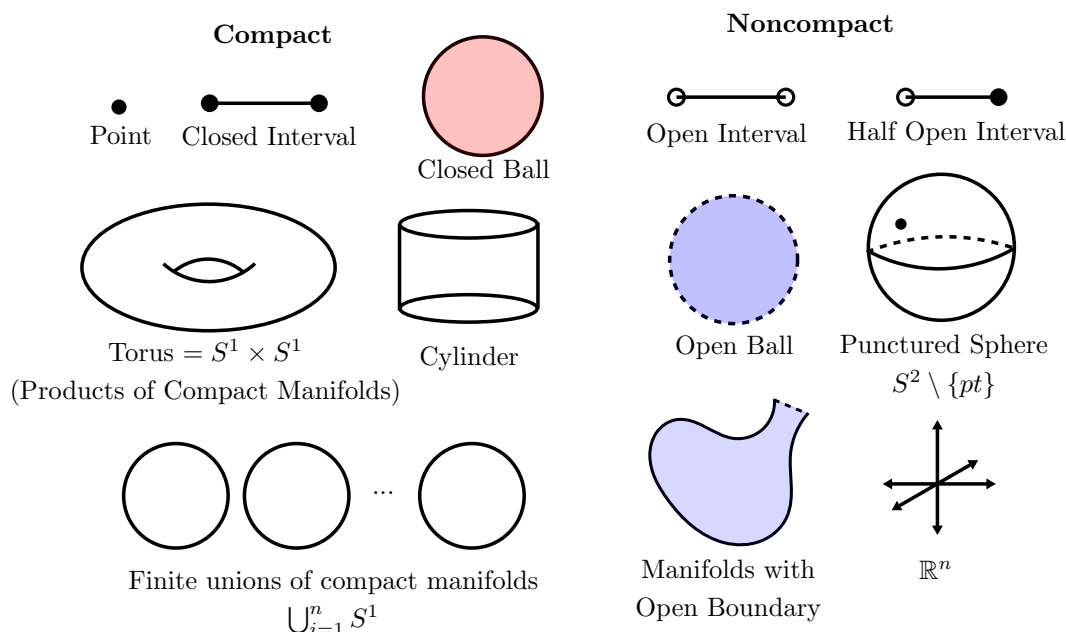


Figure 2.4: Some compact and noncompact manifolds

Compactness is essential in limiting the scope of the manifolds we're hoping to classify. Even in dimension 2, the classification of noncompact surfaces is fairly complicated. If the dimension is at least 4, this problem is harder than the word problem for groups, which is unsolvable in general. In order to avoid hopeless situations, we frequently limit our scope to connected, compact manifolds.

## 2.2 Manifolds with Boundary

Some of the examples given in the previous section were disingenuous. Until now, all of the manifolds we have seen have not had any sharp edges or endpoints, like a closed ball or closed interval. In fact, these aren't manifolds at all! Given our definition, a circle is a manifold, but a filled-in disk or a cylinder is not. This is because points on the boundary don't have neighborhoods which are diffeomorphic to open subsets of Euclidean space. However, these objects are still interesting, and they're pretty close to being manifolds

already. If we want to include them, all we have to do is generalize our definition of “manifold” a bit. Instead of using  $\mathbb{R}^k$  as our local model space, we can use the upper half space  $H^k$  consisting of all points in  $\mathbb{R}^k$  with nonnegative final coordinate. The boundary of  $H^k$  is  $\mathbb{R}^{k-1}$ , and open sets which intersect this boundary will be the local picture for neighborhoods around boundary points.

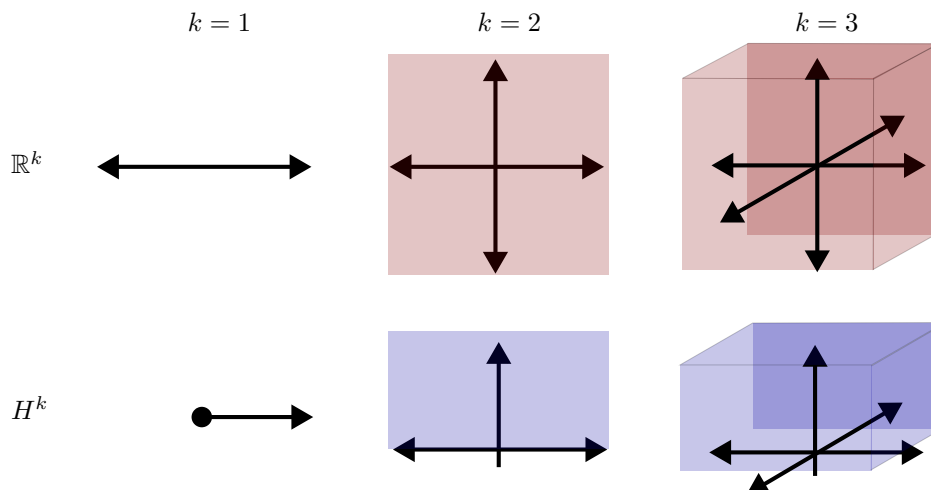


Figure 2.5:  $\mathbb{R}^k$  and  $H^k$  for  $k = 1, 2, 3$

**Definition 2.4.** A subset  $X$  of  $\mathbb{R}^N$  is a  $k$ -dimensional **manifold with boundary** if every point of  $X$  has a neighborhood diffeomorphic to an open subset in the space  $H^k$ . The **boundary** of  $X$ , denoted  $\partial X$ , consists of those points that belong to the image of the boundary of  $H^k$  under some local parametrization. Its complement is called the **interior** of  $X$ , denoted  $\text{int}(X) = X \setminus \partial X$ .

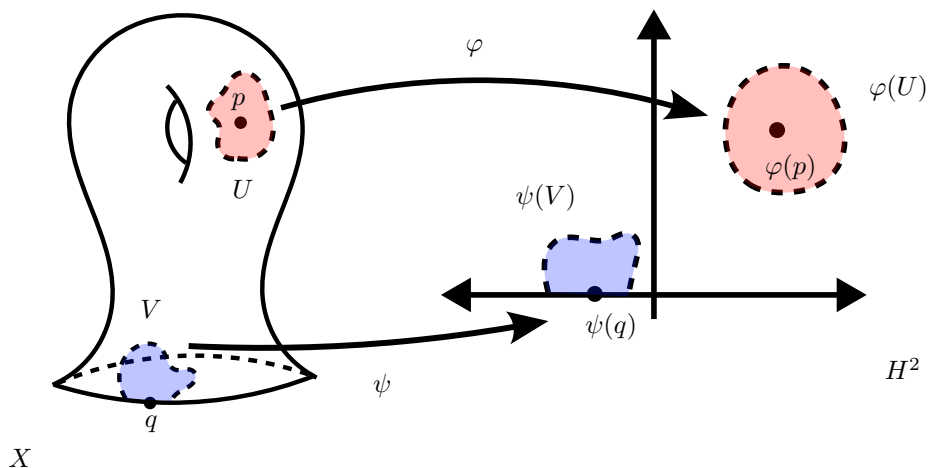


Figure 2.6: A surface with boundary  $X$  with coordinate systems  $(U, \varphi)$  and  $(V, \psi)$  around  $p$  and  $q$

**Exercise 2.5.** Prove that the closed disk

$$D^2 = \{(x, y) : x^2 + y^2 \leq 1\}$$

is a manifold with boundary. Explicitly give coordinate charts to  $H^2$  that cover every point. What is the minimum number of charts you need to do this?

There are two definitions that I need to give now that will make some of you angry. They confused me quite a bit too when I first encountered them:

**Definition 2.5.** We say a manifold  $X$  is **closed** if it is compact and its boundary  $\partial X$  is empty. A manifold  $X$  is **open** if it is noncompact and its boundary  $\partial X$  is empty.

You may notice that these definitions have *nothing* to do with the notion of open and closed sets, beyond the fact that a closed manifold is also a closed subset of  $\mathbb{R}^N$ . These terms are meant to capture some intuitive notion of a manifold being “closed up” on itself. For example, a closed disk, despite the name, is not really a “closed up” shape in any way. A sphere, on the other hand, is kind of sealed around its interior. This intuition doesn’t always work, which is why we have formal definitions, but it’s the right way to think about it. Other examples of closed manifolds include any 0-manifold, the circle, the torus, etc. Oftentimes, if we wish to ignore manifolds with boundary, we will limit our scope to closed, connected manifolds.

In addition to these new notions we have for manifolds with boundary, our other properties like connectedness and compactness work exactly the same (as we gave them, they actually work on any subset of  $\mathbb{R}^n$ ), as do all of the other notions we’ve introduced like diffeomorphisms, dimension, etc. Unfortunately, one downside of the definition for these objects is that the product of two manifolds with boundary is not generally another manifold with boundary. The square  $[0, 1] \times [0, 1]$  is an example, since the corner points don’t have a neighborhood diffeomorphic to an open subset of  $H^k$ .

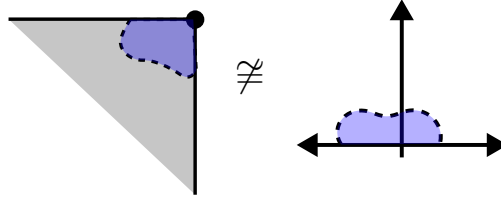


Figure 2.7: No neighborhood of the corners of a square is diffeomorphic to an open subset of  $H^2$

One thing one might think to do is to change the model space again to something like the upper half quadrant. Although this is an actual strategy that appears in the literature (see *manifolds with corners*), it will just overcomplicate our goals in these notes. Things aren’t all bad though, we do have at least one nice results with products:

**Proposition 2.1.** If  $X$  is a manifold without boundary and  $Y$  is a manifold with boundary, then  $X \times Y$  is a manifold with boundary. Furthermore,

$$\partial(X \times Y) = X \times \partial Y \quad \text{and} \quad \dim(X \times Y) = \dim X + \dim Y.$$

*Proof.* If  $U \subset \mathbb{R}^k$  and  $V \subset H^\ell$  are open, then

$$U \times V \subset \mathbb{R}^k \times H^\ell = H^{k+\ell}$$

is open. Moreover, if  $\varphi : U \rightarrow X$  and  $\psi : V \rightarrow Y$  are local parametrizations, then so is

$$\varphi \times \psi : U \times V \rightarrow X \times Y.$$

□

The fundamental usefulness of the boundaries of manifolds is due to this theorem:

**Theorem 2.3.** If  $X$  is a  $k$ -dimensional manifold with boundary, then  $\partial X$  is a  $(k-1)$ -dimensional manifold without boundary.

Because of this, two manifolds with boundaries can actually be *glued* together along their boundaries. If this is done correctly, the result is also a manifold. Similarly, two boundaries of a single manifold can be glued together.

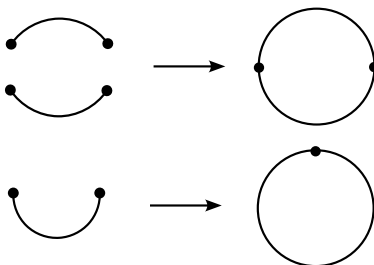


Figure 2.8: Two ways of gluing closed intervals together to get a circle

The formalism for this is a bit confusing if you haven't seen it before, and you can understand most of the rest of these notes without it. However, I will state the idea for completeness. We glue together manifolds *along diffeomorphisms of their boundaries*: if  $X$  and  $Y$  are manifolds and  $f : \partial X \rightarrow \partial Y$  is a diffeomorphism, then we have a natural way of gluing  $X$  to  $Y$ . All we do is take any point  $p \in \partial X$  and glue it to  $f(p)$ . Since  $f$  is a diffeomorphism, everything is done smoothly, and we get another manifold which we denote  $X \cup_f Y$ .

From now on, the term “manifold” will still refer to a manifold without boundary — I will do my best to specify when I mean for a manifold to have a boundary. Armed with the most general definitions we will need for the remainder of these notes, we can begin to understand the most basic instances of these objects, manifolds of dimension 0 and 1.

## 2.3 The Classification of 0- and 1-Dimensional Manifolds

We begin by classifying all 0-dimensional manifolds. Some call the study of such objects “combinatorics”.

**Theorem 2.4.** If  $X$  is a 0-dimensional manifold, then  $X$  is the union of discrete points in  $\mathbb{R}^N$ . So,  $X$  is determined up to diffeomorphism by its cardinality. Here, discrete just means there exists a neighborhood around each point which is disjoint from every other point.

*Proof.* Suppose  $X$  is a 0-dimensional manifold. Then around each point  $p \in X$ , there is a neighborhood  $U$  around  $p$  such that  $U$  is diffeomorphic to  $\mathbb{R}^0$ , which consists of a single point. Since this is the case, we see that the points in  $X$  are discretely separated. If we have another 0-manifold  $Y$  with the same cardinality as  $X$ , then choose any bijection from  $X$  to  $Y$ . This map turns out to be a diffeomorphism, since it's smooth with smooth inverse (both maps are locally constant, so all partial derivatives vanish).  $\square$

**Exercise 2.6.** Show that a 0-manifold is compact if and only if its cardinality is finite.

This completes the classification of all 0-manifolds, with or without boundary (since those two ideas are the same in this case). Now we come to a more interesting theorem, the classification of 1-manifolds with boundary.

**Theorem 2.5.** Let  $X$  be a connected 1-dimensional manifold with or without boundary. Then  $X$  is diffeomorphic to  $\mathbb{R}$ , the half open interval  $[0, \infty)$ , the closed interval  $[0, 1]$ , or  $S^1$ . If  $X$  is disconnected, then it is the union of disjoint copies of these four manifolds.



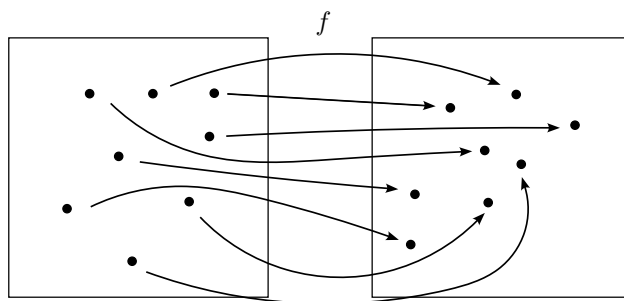


Figure 2.9: Two 0-manifolds with the same cardinality are diffeomorphic

The proof of this theorem is nontrivial. There are a number of proof techniques I like, for instance those in Dan Freed’s notes on *Differential Topology* (<https://people.math.harvard.edu/~dafr/difftop.pdf>) or Mike Hopkin’s notes [here](#). The first uses the principle of introducing some local geometric structure on a manifold and using it to prove a global result. The latter is my preferred method, given by introducing a *Morse function*  $f : X \rightarrow \mathbb{R}$ , namely a function with finitely many nondegenerate critical points (meaning the second derivative of  $f$  at a critical point is nonzero). Simple surgeries are used to eliminate most or all of the critical points, once again giving us control over the global structure of  $X$ . Note the duality of these two approaches: in the first, we map the “standard” manifold  $\mathbb{R}$  into our space, and in the second, we map our space onto the standard model. We will explore Morse theory more in depth later in these notes.

**Exercise 2.7.** In addition to Exercise 1.3, the following might help you understand why these are the only things we can get in 1 dimension. Let  $a < b$  be real numbers. Prove that any open interval  $(a, b)$ , half open interval  $[a, b)$  or  $(a, b]$ , or closed interval  $[a, b]$  is diffeomorphic to  $\mathbb{R}$ ,  $[0, \infty)$ , or  $[0, 1]$ , respectively. You should construct explicit diffeomorphisms for each (some may be similar). Remember, because of Exercise 1.6, you can apply a few diffeomorphisms in a row to make things easier for yourself.

### 3 Surfaces

Now that we know everything there is to know about dimensions 0 and 1, we can shift our focus to dimension 2. Already, things become *much* more interesting. We will be working our way up to the classification of closed connected surfaces in future sections, but for now, let's just have some fun with paper and glue.

#### 3.1 Building Surfaces

Say we start with a piece of origami paper. In the following few examples, I will denote the edges we're gluing together with the same color. If you can, try to think about the questions posed before looking at the answers.

- (1) If I take the left and right edge of the origami paper and glue them together (counter to the point of origami), what shape do I get?

Indeed, I get a cylinder:

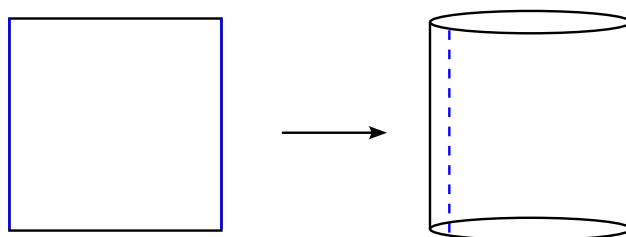


Figure 3.1: Gluing paper to get a cylinder

A cylinder is a classic example of a manifold with boundary, where in this case the boundary is the union of two disjoint circles.

- (2) Now, what happens if I take those two circles in the boundary, and I glue them together by simply stretching out the cylinder when necessary and match up points that have the same  $x$  and  $y$  coordinates? (the  $z$ -axis is up and down in the picture below)

We get a torus!

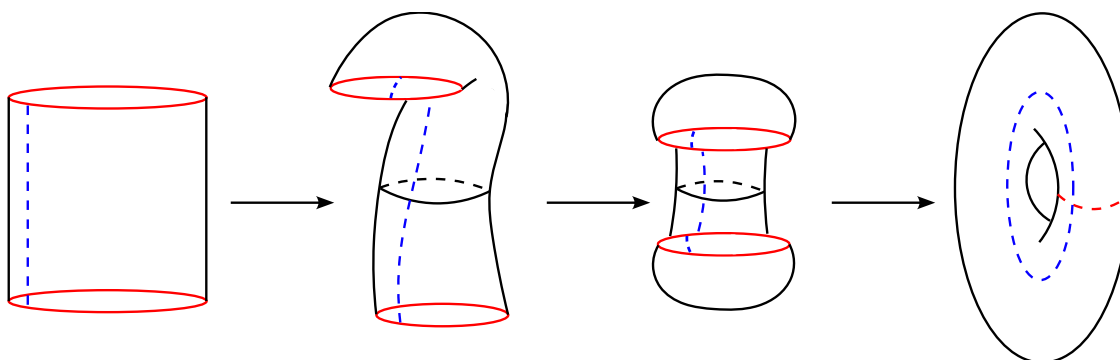


Figure 3.2: Gluing a cylinder together to get a torus

So, all in all, we went from a piece of origami paper to a torus (a surface without boundary) by this sequence of gluing, which we might visualize as so:

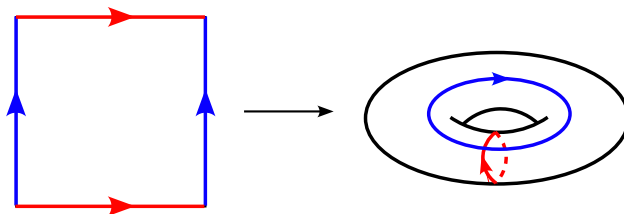


Figure 3.3: Gluing a square into a torus

In the figure, we use arrows as instructions for how to glue together the edges. Formally speaking, if the arrows are in the same direction, we glue along the *identity* diffeomorphism, namely the map sending the “same” points (ignoring their relative position in  $\mathbb{R}^N$ ) in either boundary component to each other. If they are facing opposite directions, we glue along *negative* the identity, where we first flip and then match up the points.

- (3) What if in the first step, we glued with the opposite orientation? That is, if we flipped the left or right edge before we glued them together — what shape would we get?

We get a Möbius band!

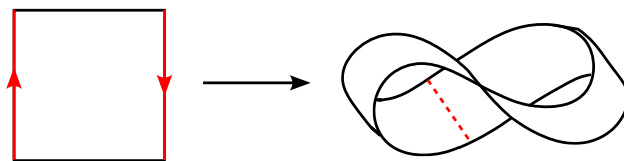


Figure 3.4: Gluing together a Möbius band

**Exercise 3.1.** A Möbius band is an example of a surface with boundary. What is its boundary? Identify it in both the gluing diagram and the final picture.

**\*Exercise 3.2.** What happens if we now glue the remaining boundary to itself via the identity (without a flip) on the two parts shown in the gluing diagram (top and bottom edge)? Is this a manifold? Have you seen anything similar to it before?

- (4) Of course, there’s no reason we had to start with a square. If we instead start with some other polygon, we can get different surfaces. So long as we glue each edge to another edge, the result will be a closed surface. With this particular gluing pattern, we get something that looks like the surface of a doughnut with two holes:

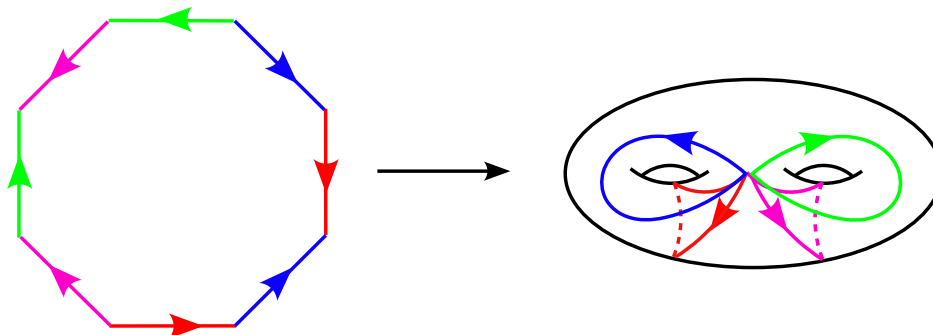


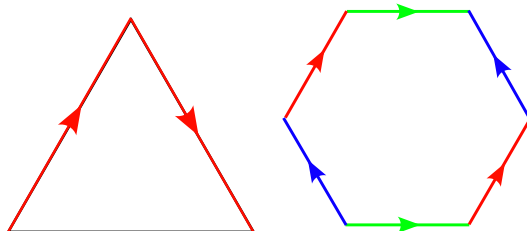
Figure 3.5: Gluing together an octagon to get a genus 2 surface

We call it a **genus 2 surface**. As you might imagine, the **genus** of such a surface is the number of holes it has.

**Exercise 3.3.** Draw pictures at each step of the gluing — prove that the picture I drew is correct.

Now, these exercises are how I want you to spend most of your time in this section:

**Exercise 3.4.** What shapes do we get if we perform the following gluings? Make sure to draw where the edges end up after the gluing.



**Exercise 3.5.** Find the minimum  $n$  such that we can glue together a genus  $g$  surface using an  $n$ -gon (a polygon with  $n$  edges).

**\*Exercise 3.6.** Prove your answer.

Feel free to play with the gluing orientations to see what wacky shapes you can come up with. Many of them won't be possible to see using only 3 dimensions,<sup>8</sup> so don't be worried if things don't fit together quite right. An interesting question to think about then, is when *can* we see the whole of a surface?

## 3.2 Orientability

Very commonly, when working with manifolds, one has a notion of a local or *infinitesimal* structure and tries to extend it globally. In the case of closed surfaces, one such structure entirely determines whether or not it can be viewed in 3 dimensions: *orientability*. Intuitively, a surface is orientable when we are able to make a consistent choice for clockwise and counterclockwise. Consistent in this case just means that, no matter how we walk around the surface, clockwise stays clockwise and counterclockwise stays counterclockwise. We formalize this using the language of *loops*.

**Definition 3.1.** A **loop** on a manifold  $X$  is a path  $\gamma : [0, 1] \rightarrow X$  such that  $\gamma(0) = \gamma(1)$ . Equivalently, a **loop** is a smooth map  $\gamma : S^1 \rightarrow X$ , where  $S^1$  denotes the circle.

As I mentioned, orientation is a local property. This means that we need a definition of “clockwise” and “counterclockwise” at each point of a manifold that we can check is consistent along loops. How do we make this precise? I am going to take an uncommon approach to this definition so that I can avoid introducing unnecessary terminology and concepts. What we'll do here is use that there's really only two possible orientations at any point, and they're basically the same up to a reflection:

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<sup>8</sup>One can actually prove that all surfaces can be *embedded* in 4 dimensions, so if we were 4D creatures we could see them in their entirety.

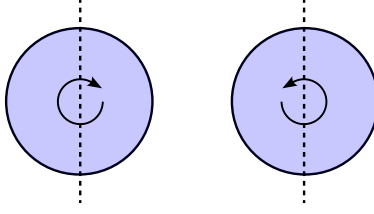


Figure 3.6: The two different orientations we can have in a local picture

**Definition 3.2.** Let  $X$  be a surface and let  $p$  be some point in  $\mathbb{R}^2$ . Consider the ball  $B_1(p)$ . Given a loop  $\gamma$  in  $X$ , only one of the following can be true:

- (1) There is a smooth family of diffeomorphisms  $\Gamma_t$  from the ball  $B_1(p)$  into  $X$  such that  $\Gamma_t(p) = \gamma(t)$  for all  $t \in [0, 1]$  and  $\Gamma_1 = \Gamma_0$ . In this case, we call  $\gamma$  a **orientation-preserving loop**.
- (2) There is a smooth family of diffeomorphisms  $\Gamma_t$  from the ball  $B_1(p)$  into  $X$  such that  $\Gamma_t(p) = \gamma(t)$  for all  $t \in [0, 1]$  and  $\Gamma_1 = \Gamma_0 \circ R$ , where  $R$  is some reflection of  $B_1(p)$ .<sup>9</sup> In this case, we call  $\gamma$  a **orientation-reversing loop**.

Don't think too much about the phrase “smooth family of diffeomorphisms”, your intuition about moving around a small disk centered at points on the loop and seeing if clockwise changed to counterclockwise within that disk is essentially correct.

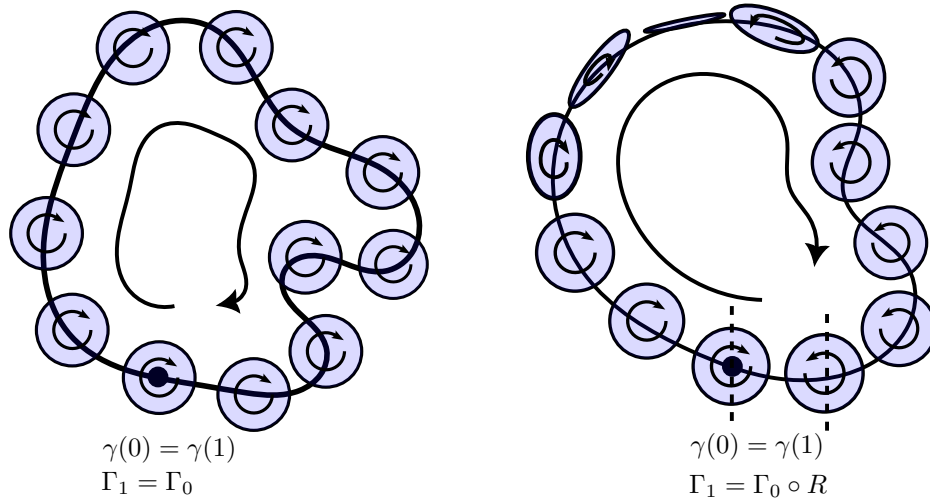


Figure 3.7: An orientation-preserving and orientation-reversing loop

We say that  $X$  is **orientable** if there does not exist an orientation-reversing loop in  $X$ . If  $X$  is orientable, a choice of either clockwise or counterclockwise is called an **orientation**.

Note, this definition is *very* close to generalizing to arbitrary  $k$ -manifolds. Loops work exactly the same way, all we would need to do is consider what an orientation means in higher dimensional balls (so that we can define it for the local picture). Let me know if you want to hear more about this, I might put it in an appendix somewhere.

Now, to get a handle on the idea of (non-)orientability, let's consider the paragon example, the Möbius band. It's pretty straightforward to imagine an orientation-reversing loop, just take the “equator”.

<sup>9</sup>Maybe you can prove that a reflection is a diffeomorphism.

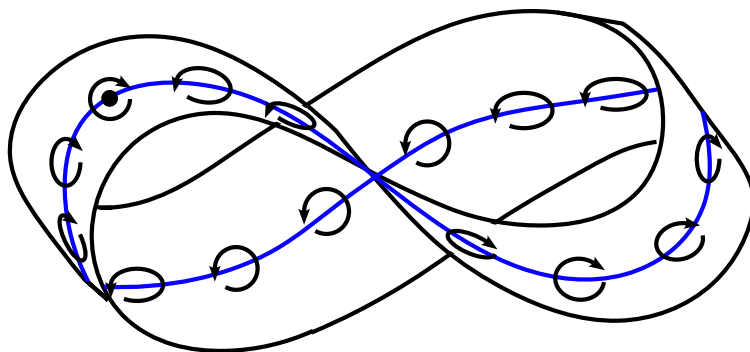


Figure 3.8: The Möbius band is not orientable

An observation that I find really cool is that the definition I gave for orientable is equivalent to a different, wackier one:

**Definition.** A surface  $X$  is **orientable** if it doesn't have a Möbius band as a submanifold.

**\*Exercise 3.7.** Prove that these definitions are equivalent.

Other examples of non-orientable surfaces include the *Klein bottle* and the *real projective plane*, or  $\mathbb{RP}^2$  for short.

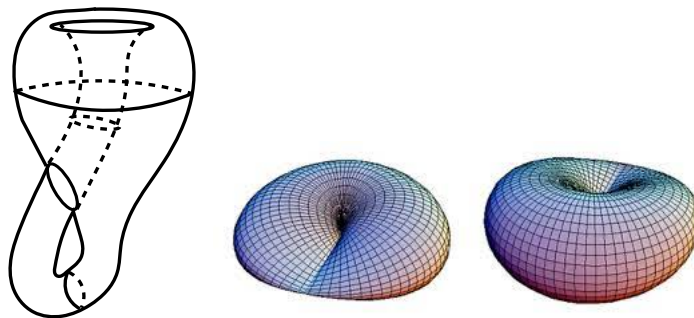


Figure 3.9: Representations of the Klein bottle and real projective plane

These are surfaces who cannot be seen in their entirety within  $\mathbb{R}^3$ ; we would need 4-dimensional brains to see them without any self intersections. Especially with  $\mathbb{RP}^2$ , it can be difficult to conceptualize what one is looking at in these pictures. One fairly efficient way to understand these objects is from their gluing diagrams (which you might identify as the solutions to Exercise 3.2). In fact, we can easily see the Möbius bands embedded in the surfaces using these diagrams.

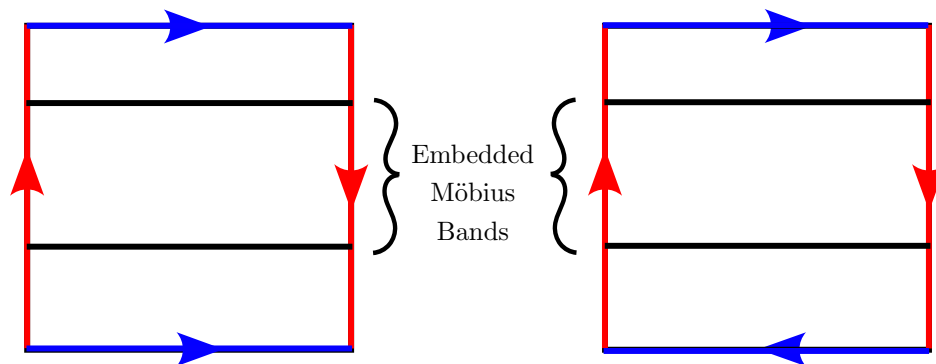


Figure 3.10: Gluing diagrams for the Klein bottle and real projective plane

After the gluing, these highlighted regions look just like a rectangle with a pair of opposite edges glued together with a twist, i.e. the Möbius band. If you want a challenge, try to locate these Möbius bands in the 3D representations of the Klein bottle and  $\mathbb{RP}^2$  shown above.

All of the other surfaces we have seen so far have been orientable. You may have noticed that all of the *closed* surfaces have essentially just been built around either the torus (in the case of genus  $g$  surfaces) or around the Möbius band. In fact, this is the beginning observation that will lead to the classification of closed surfaces.

## 4 Classification of Surfaces

Now that we’ve played with surfaces for a bit and have some understanding of how they work, we can start working towards their classification. In this section, I will introduce the language in which we will perform this classification, as well as the method of its proof.

### 4.1 Connected Sums

The classification of surfaces relies on the notion of a *decomposition* into simpler surfaces, and in order to state this, we need to know how we can combine those pieces in the first place. In previous sections, we have seen a few ways of taking two manifolds and producing a new one: Cartesian products, disjoint unions, and gluing. The first of these three, in general, outputs a manifold of different dimension than its two inputs — no good. The second is a great way of producing disconnected manifolds that aren’t particularly interesting to us, so that is no good either. The method I will demonstrate here is a particularly nice instance of *gluing* for surfaces.

**Definition 4.1.** Let  $X$  and  $Y$  be two connected surfaces, potentially with boundary. In both surfaces, identify an open disk in the interior of the surface and remove it. This creates a boundary component diffeomorphic to  $S^1$  in both  $X$  and  $Y$ . We form the **connected sum**  $X \# Y$  of  $X$  and  $Y$  by gluing them along this now common boundary.

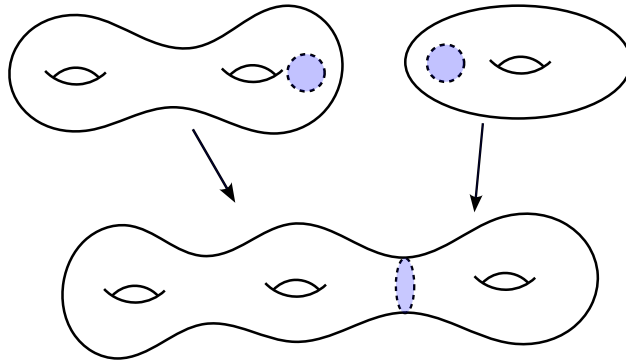


Figure 4.1: Connected sum of a genus 2 and genus 1 surface is a genus 3 surface

Now, as usual, it’s important we verify that this operation is actually well defined.

**Theorem 4.1.** Up to diffeomorphism, the connected sum  $X \# Y$  of two connected surfaces  $X$  and  $Y$  is independent of the choice of disks in  $X$  and  $Y$ .

*Proof.* First, observe that the connected sum is a *commutative* operation, namely  $X \# Y \cong Y \# X$  (what is this diffeomorphism?). So, it is enough to show that if we choose disks  $D$  and  $D'$  in  $X$ , there is a diffeomorphism  $X \setminus D \rightarrow X \setminus D'$ . Since  $X$  is connected, it is also path connected. Let  $p$  and  $q$  be the centers of  $D$  and  $D'$ , and let  $\gamma : [0, 1] \rightarrow X$  be a path with  $\gamma(0) = p$  and  $\gamma(1) = q$ . There is a theorem in differential topology called the *tubular neighborhood theorem*. The details aren’t terribly important for our purposes, but in essence this theorem says that for any submanifold  $Z$ , there is an open neighborhood  $U$  containing  $Z$  which “hugs” the submanifold as closely as we want (given some  $\epsilon > 0$ , there is a neighborhood  $U$  such that the distance from the any point<sup>10</sup>  $p \in \partial Z$  to the boundary of  $U$  in the normal direction away from  $Z$  is  $\epsilon$ ).

<sup>10</sup>If  $\partial Z$  is empty, just use points in  $Z$  itself



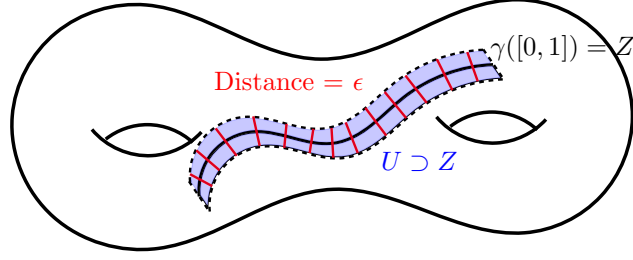


Figure 4.2: The tubular neighborhood of a curve  $\gamma$  in a genus 2 surface

Assuming this theorem is true,<sup>11</sup> we can apply it to get a tubular neighborhood  $U$  around the image of  $\gamma$ . Since  $U$  is open, it must contain small disks  $\Delta$  and  $\Delta'$  around  $p$  and  $q$  respectively. Now we can begin constructing our diffeomorphism; we do so by composing three simpler ones.

- (1) Since  $D$  and  $\Delta$  share the same midpoint  $p$ , let  $f : X \setminus D \xrightarrow{\sim} X \setminus \Delta$  be the map which simply shrinks  $D$  onto  $\Delta$ .

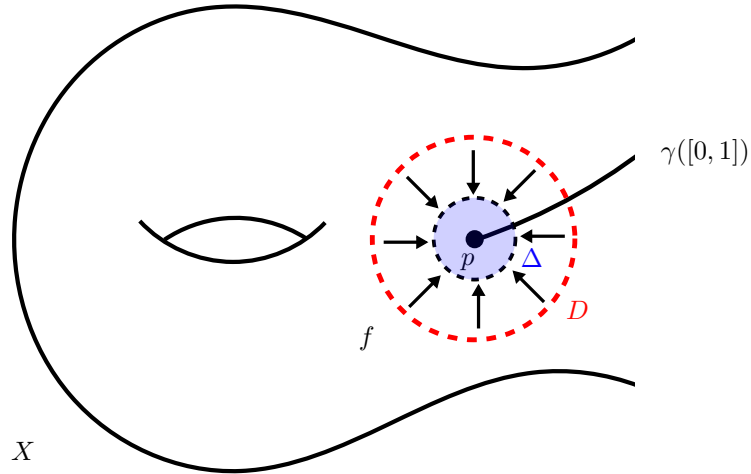


Figure 4.3: Visualization of  $f$  shrinking  $D$  onto  $\Delta$

- (2) Using our neighborhood  $U$ , transport  $\Delta$  to  $\Delta'$  by “sliding” it through  $U$ . This will give us a diffeomorphism  $g : X \setminus \Delta \xrightarrow{\sim} X \setminus \Delta'$  — here’s how it works intuitively. Outside of  $U$ , let  $g$  be the identity. As we slide  $\Delta$  to  $\Delta'$ , think of a hockey puck with 1-dimensional “springs” attached to it on either side, representing copies of the image of  $\gamma$  within  $U$ . As we slide  $\Delta$ , the springs behind it expand and get stretched out while the springs after it contract and get bunched up. Keep track of how these points expand and contract, and let this be how  $g$  acts on  $U$ .

<sup>11</sup>If you’d like to see a proof, you should consider taking a class on differential topology!

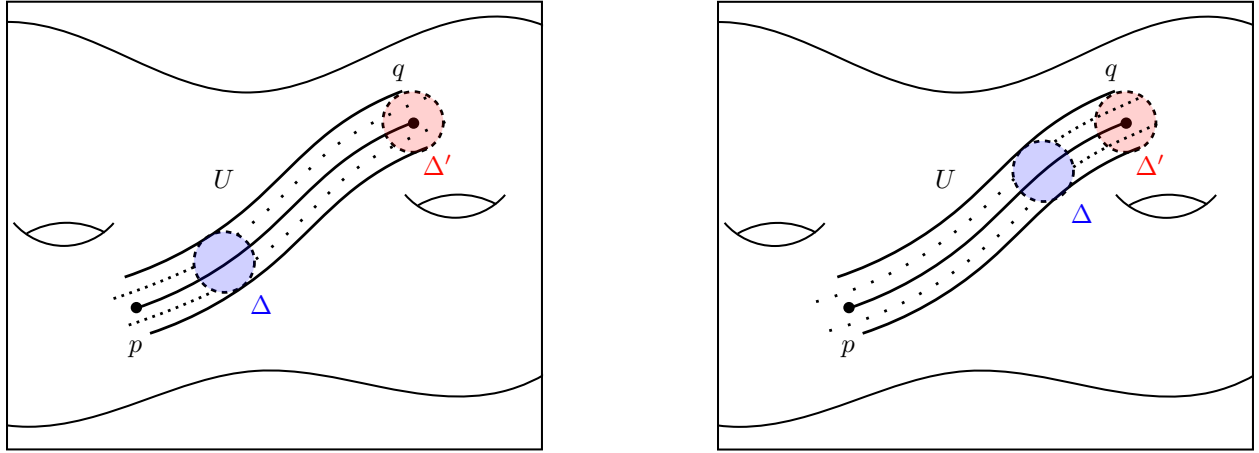


Figure 4.4: Sliding  $\Delta$  to  $\Delta'$  and keeping track of the “springs”

(3) Now, to finish off, just expand  $\Delta'$  back into  $D'$  to get a diffeomorphism  $h : X \setminus \Delta' \rightarrow X \setminus D'$ .

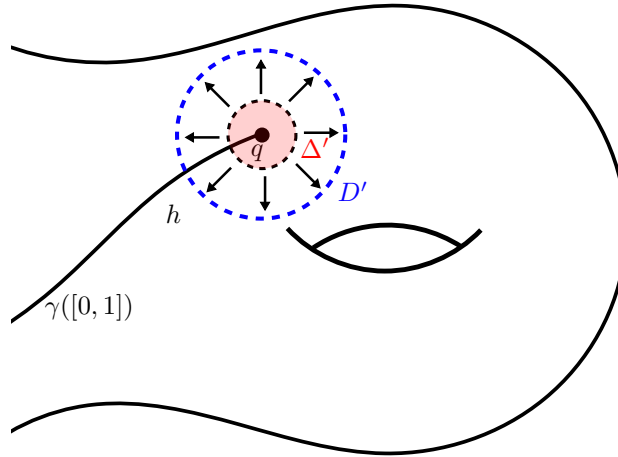


Figure 4.5: Expanding  $\Delta'$  into  $D'$  via  $h$

We get a final diffeomorphism from  $X \setminus D \xrightarrow{\sim} X \setminus D'$ , as desired:

$$\begin{array}{ccccccc}
 X \setminus D & \xrightarrow[\sim]{f} & X \setminus \Delta & \xrightarrow[\sim]{g} & X \setminus \Delta' & \xrightarrow[\sim]{h} & X \setminus D' \\
 & & & & \searrow \scriptstyle h \circ g \circ f & & \\
 & & & & \sim & & 
 \end{array}$$

□

**Exercise 4.1.** Now that we’re convinced this operation is legit, let’s see some results on the properties we’ve seen so far. Let  $X$  and  $Y$  be connected surfaces.

(1) Prove that  $X \# Y$  is connected.

- (2) Prove that  $X$  and  $Y$  are compact if and only if  $X \# Y$  is compact. Since the connected sum doesn't introduce any more boundary components, this implies that  $X$  and  $Y$  are closed if and only if  $X \# Y$  is closed.

Although one can of course visualize the actual surfaces when performing a connected sum, it is often more useful to again use gluing diagrams. Connected sums have a very neat and tidy representation with these diagrams: suppose you have two polygons  $P$  and  $Q$  which you're gluing into closed surfaces  $X$  and  $Y$  (does this give any restrictions on the number of edges in  $P$  and  $Q$ ?). In this scenario, all of the vertices of one of your polygons are glued together into a single point (prove this if you are not convinced). So, pick two vertices in  $P$  and  $Q$  and "extend" them into new edges, say  $e_P$  and  $e_Q$ . In the final gluing, the addition of these edges takes what was some point on the surface and opens it up into a circle boundary component, which is the same thing as removing a disk centered at that point. Now, we can safely glue together  $e_P$  and  $e_Q$  to get another polygon gluing together to get a closed surface, namely one diffeomorphic to  $X \# Y$ .

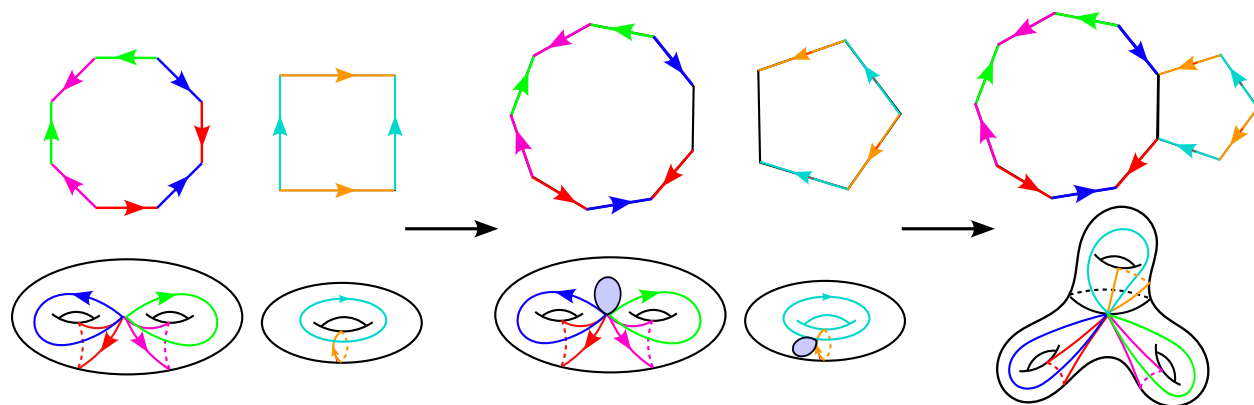


Figure 4.6: Connecting the gluing diagrams for a genus 2 surface  $X$  and genus 1 surface  $Y$  yields a diagram for a genus 3 surface  $X \# Y$

This may be helpful as an approach to Exercise 3.6.

## 4.2 Cell Structures and Euler Characteristic

The property we will be exploring in this section is one of the first that could genuinely be considered a *topological invariant*, dating all the way back to Euler (as the name suggests). An *invariant* is a relatively simple but *very* important concept in mathematics. An invariant is a property of some collection of mathematical objects such that objects which are “the same” in some sense have the same value for this property. In topology, we care about invariants up to continuous deformation. If you have taken a class on discrete math or something else involving graph theory, you may have seen the *Euler characteristic* defined for a connected planar graph as

$$\chi = V - E + F$$

where  $V$  is the number of vertices,  $E$  the number of edges, and  $F$  the number of faces in the graph (including the exterior face). Alternatively, you may have seen it defined for the surface of a polyhedra, where you treat its vertices and edges a graph and set  $F$  to the number of its faces. Euler proved in 1758 that for polyhedra,

$$\chi = V - E + F = 2.$$

This is the first indication of something topological going on — here,  $\chi = 2$  turns out to be the Euler characteristic of the sphere! We will define *Euler characteristic* for any surface, and, more broadly, any  $k$ -manifold.

Before we get to this definition, we need to discuss a broad generalization of the examples above.

**Definition 4.2.** Let  $X$  be an  $n$ -manifold (potentially with boundary). A **cell structure** or **cell decomposition** on  $X$  is a decomposition<sup>12</sup> of  $X$  into subsets

$$X_0 \subset X_1 \subset \cdots \subset X_n = X$$

such that  $X_k$  is a union of  $k$ -dimensional closed balls for  $0 \leq k \leq n$ . We call these balls  **$k$ -cells** and  $X_k$  the  **$k$ -skeleton** of  $X$ .

You'll notice that if  $X$  is a manifold with boundary, a cell structure on  $X$  (of dimension  $n$ ) necessarily induces a cell structure on  $\partial X$  (of dimension  $n - 1$ ).

Although this definition may seem quite complicated at first, it becomes fairly intuitive with some practice. All that we're doing here is turning  $X$  into a  $k$ -dimensional “graph”. In fact, the set  $X_1$  in any cell decomposition is literally a graph embedded in  $X$ . Here is a classic cell structure on the sphere:

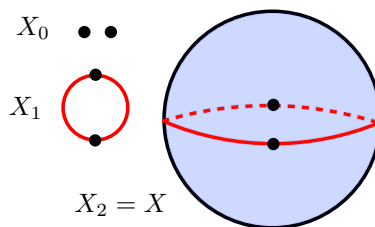


Figure 4.7: A cell decomposition of  $S^2$ : vertices are 0-cells, edges are 1-cells, and faces are 2-cells

If we added a 3-cell inside of the sphere, we would have a cell decomposition for a closed ball. For surfaces, cell structures are pretty straightforward: just draw a graph on the surface such that the faces formed by the graph are diffeomorphic to disks. Another way to think of this condition is, if you were to cut along the graph, you would get a union of a bunch of disjoint disks.

**Exercise 4.2.** Put a cell structure on a circle, a disk, a torus, and a genus 2 surface. Can you relate the latter two in any way to their gluing diagrams?

**\*Exercise 4.3.** Put a cell structure on a *genus 2 handle*, that is a “filled-in” genus 2 surface. This will require some 3-cells, since it’s a 3-manifold with boundary.

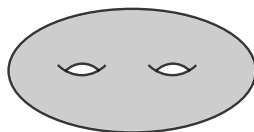


Figure 4.8: A genus 2 handle

**\*Exercise 4.4.** Put a cell structure on  $\mathbb{RP}^2$ .

Now we arrive at our definition:

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<sup>12</sup>Some might call this a *filtration*.

**Definition 4.3.** Let  $X$  be a compact  $n$ -manifold and  $X_0 \subset X_1 \subset \cdots \subset X_n = X$  a cell decomposition with  $b_k$   $k$ -cells in each  $k$ -skeleton. The **Euler characteristic** of  $X$  is given by the alternating sum

$$\chi(X) = b_0 - b_1 + b_2 - b_3 + \dots$$

**Exercise 4.5.** Why do we require compactness in this definition?

**\*Exercise 4.6.** Prove your answer.

Although this invariant is more subtle than previous topological properties we've discussed, we have that

**Theorem 4.2.** If  $X$  and  $Y$  are diffeomorphic manifolds, then  $\chi(X) = \chi(Y)$ . Alternatively, if  $\chi(X) \neq \chi(Y)$ , then  $X$  and  $Y$  are not diffeomorphic.

Without additional notions coming from algebraic topology (specifically *homology*), it is surprisingly difficult to prove this theorem. If we have time, we might talk about this proof in more detail later.

**Exercise 4.7.** Let  $X$  be a genus  $g$  surface. Compute  $\chi(X)$  in terms of  $g$ .

**Exercise 4.8.** If  $X$  and  $Y$  are connected surfaces, prove that  $\chi(X \# Y) = \chi(X) + \chi(Y) - 2$ .

## 5 Classification of Surfaces Cont.

With this setup, we are finally ready to see one of the many, *many* proofs of the classification of surfaces.

**Theorem 5.1.** If  $X$  is a closed, connected, orientable surface, then  $X$  is diffeomorphic to a genus  $g$  surface. In other words, if  $\Sigma$  is the torus, then  $X = \underbrace{\Sigma \# \cdots \# \Sigma}_{g \text{ times}}$ .

### 5.1 Proof

This proof is from a paper by Andrew Putman, but the argument is originally due to Christopher Zeeman. Let  $X$  be a closed connected orientable surface. We begin with a lemma:<sup>13</sup>

**Lemma 1.** If  $\chi(X) = 2$ , then  $X$  is diffeomorphic to  $S^2$ . This will additionally prove that  $\chi(X) \leq 2$  for all connected compact surfaces  $X$  (including non-orientable and with boundary!).

*Proof.* Let  $X$  be any connected compact surface. Fix a cell decomposition  $X_0 \subset X_1 \subset X_2 = X$ . Our first goal is to somehow relate the Euler characteristic of  $X$  to this cell decomposition. So, choose a *spanning tree*  $T$  in the 1-skeleton of  $X$ . A spanning tree is simply some collection of edges (a connected subgraph) in  $X_1$  that touches every vertex in  $X_1$  (maybe you can prove such a thing exists). Let  $\Gamma$  be the graph with one vertex in each 2-cell of  $X$  where two vertices are connected by an edge exactly when their associated 2-cells share an edge of the cell structure which does not lie in  $T$ .

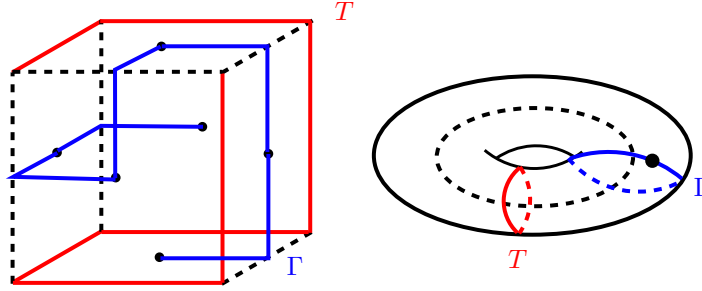


Figure 5.1: Examples of  $T$  and its *dual graph*  $\Gamma$

In the figure, we intend for the first example to have  $X_0, X_1$ , and  $X_2$  as the usual vertices, edges, and faces of a cube, and the second example to have one vertex in  $X_0$ , two edges forming loops in  $X_1$ , and a single face in  $X_2$  forming the surface of a torus (why is this diffeomorphic to a disk?). Admittedly, the first example in the figure is not on a smooth manifold, but it lends some intuition for how to think about these graphs (if you wanted, you could *smooth out* the faces in such a way that they meet smoothly at the edges and vertices — you’ll find that you get a sphere). Suppose that our cell structure has  $V$  vertices,  $E$  edges, and  $F$  faces. Since  $T$  spans  $X_1$ , it must have  $V$  vertices and  $E_0$  edges for some  $E_0 \leq E$ . Also, by construction, the graph  $\Gamma$  must have  $F$  vertices and  $E_1$  edges for some  $E_1 \leq E$ . Note,  $E = E_0 + E_1$ , since the edges of  $\Gamma$  precisely correspond to the edges “missing” from  $T$ . Therefore,

$$\chi(X) = V - E + F = (V - E_0) + (F - E_1) = \chi(T) + \chi(\Gamma)$$

where we define  $\chi(T)$  and  $\chi(\Gamma)$  to be the Euler characteristics of  $T$  and  $\Gamma$  as graphs.

Now that we have a relationship between  $\chi(X)$  and  $X$ ’s cell decomposition, we can make some clever observations to conclude the lemma.

<sup>13</sup>This is the first of a few “*Poincaré Conjectures*” we will explore.

**Exercise 5.1.** Let  $G$  be a connected graph with  $v$  vertices and  $e$  edges. Then  $\chi(G) = v - e \leq 1$ , and moreover,  $\chi(G) = 1$  if and only if  $G$  is a tree.

From the exercise, we find that  $\chi(T) = 1$  and

$$\chi(X) = 1 + \chi(\Gamma) \leq 2.$$

Moreover,  $\chi(X) = 2$  if and only if  $\Gamma$  is a tree. Suppose that this is true, and  $\Gamma$  is a tree. We want to show that  $X \cong S^2$ . We can find disjoint, closed neighborhoods  $U$  and  $V$  around  $T$  and  $\Gamma$ , respectively, such that  $U$  and  $V$  are diffeomorphic to disks.

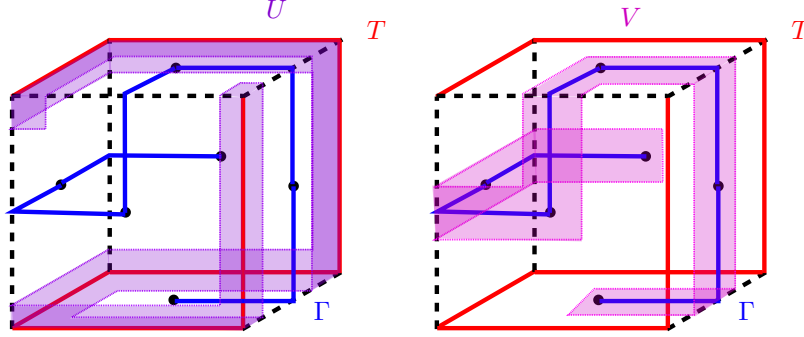


Figure 5.2: Closed neighborhoods  $U$  and  $V$  around  $T$  and  $\Gamma$ , respectively (again, up to some smoothing)

$U$  and  $V$  are disks because both  $T$  and  $\Gamma$  are trees, and they are disjoint by the relative construction of  $T$  and  $\Gamma$ . Let  $U$  and  $V$  expand outwards until they touch along their boundaries. This gives a decomposition of  $X$  into two disks glued along their common boundary circle, hence  $X$  is diffeomorphic to the sphere.<sup>14</sup>  $\square$

Now, we are ready to tackle classification.

*Proof of Theorem 5.1.* Suppose that  $X$  has  $\chi(X) < 2$ . We proceed by induction, where the lemma above gives us our base case. Suppose for induction that if  $Z$  is a surface with even Euler characteristic  $\chi(Z)$  which is strictly less than  $\chi(X)$ , then  $Z$  is diffeomorphic to a genus  $g - 1$  surface for some  $g \geq 1$ . We want to show that  $X$  is diffeomorphic to a genus  $g$  surface. So, let  $T$  and  $\Gamma$  be as in the proof of the lemma, so that this time,  $\Gamma$  is not a tree. In particular, this implies that  $\Gamma$  contains a loop  $\gamma$ . We claim that if we cut  $X$  along  $\gamma$ , then the result is connected. To see this, pick a point  $p$  on one side of  $\gamma$  and a point  $q$  on the opposite side. We explicitly construct a path from  $p$  to  $q$  to see that  $X \setminus \gamma$  is path connected, hence connected. Pick an edge in  $\Gamma$  such that the edge is entirely contained in  $\gamma$ . Note that by construction, this edge intersects a 1-cell  $e$  in  $X_1$  at exactly one point, so that we find exactly two pieces of  $e$  on either side of  $\gamma$ . Suppose that  $v_0, v_1 \in X_0$  are the vertices of  $e$ , where  $v_0$  and  $v_1$  are on the same side of  $\gamma$  as  $p$  and  $q$ , respectively.

<sup>14</sup>We will see this technique of “expanding out” neighborhoods of the 1-skeleton again to prove the existence of *Heegaard splittings* of 3-manifolds.

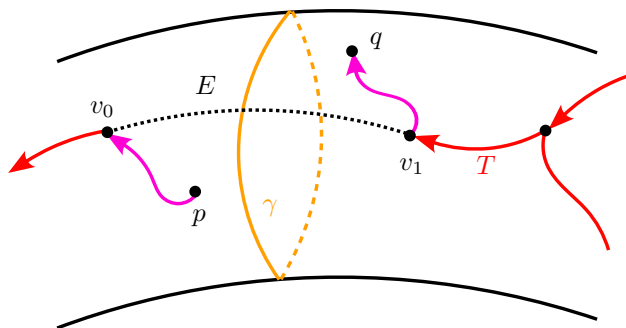


Figure 5.3: Finding a path from  $p$  to  $q$

Construct the path from  $p$  to  $q$  as follows: start by connecting  $p$  to  $v_0$ . Now, since  $T$  is a spanning tree, follow a path in  $T$  onto  $v_1$ . Now that we're on the same side as  $q$ , connect  $v_1$  to  $q$ . This path cannot cross  $\gamma$ , since we know that  $T$  is disjoint from  $\Gamma$ , hence disjoint from  $\gamma$  as well. Therefore, we find that if we remove a tubular neighborhood of  $\gamma$  from  $X$ , we get a surface with two circle boundary components. Glue two disks onto  $X$  to “cap off” the boundary components. This gives us a closed orientable surface  $Y$  with  $\chi(Y) = \chi(X) + 2$ . Figure 5.4 gives a more complicated cell structure on the torus (make sure you understand how all of these pieces fit together!) as well as the process of identifying  $\gamma$ , cutting to get two boundary components, and gluing to cap off. Now, by the inductive hypothesis, we conclude that  $Y$  is diffeomorphic to a genus  $g - 1$  surface for some  $g \geq 1$ . By construction,  $X$  is the connected sum of  $Y$  with a torus, and so  $X$  is diffeomorphic to a genus  $g$  surface, as desired.  $\square$

**Exercise 5.2.** In our proof, where did we use the fact that  $X$  is closed, connected, and/or orientable? Find the explicit steps that required each of these hypotheses — I may have purposely left out some justification at these steps.

## 5.2 Statement of the Non-Orientable Case

This concludes the proof of the classification of closed, connected, orientable surfaces. Of course, as with all classification theorems, it is interesting to see if we can generalize these results to any immediate related results. As is frequently the case in topology, these related results often come from dropping one or more adjectives in front of the objects that we're studying. What if we allowed for surfaces that are with boundary, non-orientable, or non-compact? As it turns out, all of these questions have known answers. If you are interested, make sure to look up the first one — it's relatively easy to understand.<sup>15</sup> However, I will focus on the second, since the result is so similar and fits so well with the orientable version. Here is the statement:

**Theorem 5.2.** Let  $X$  be a closed connected surface and  $\mathbb{RP}^2$  the real projective plane. Then

$$X \cong \underbrace{(\mathbb{RP}^2 \# \cdots \# \mathbb{RP}^2)}_{h \text{ times}}$$

for some  $h \geq 0$ . You can think of  $h$  as some kind of “non-orientable” genus, in analogy with the orientable case.

This theorem looks a bit weird at first. The surface  $T^2 \# \mathbb{RP}^2$  is non-orientable, but is it homeomorphic to some connect sum of  $\mathbb{RP}^2$ s?

<sup>15</sup>Since 1-manifolds are so simple, they're basically all just “punctures” of surfaces without boundary, or surfaces where we remove some disks.



**Exercise 5.3.** Prove that the Klein bottle  $K$  is diffeomorphic to  $\mathbb{RP}^2 \# \mathbb{RP}^2$ . This one is really neat I think — can you identify the surface we end up gluing to itself to get  $K$ ? The one we get when we remove a disk from  $\mathbb{RP}^2$ ?

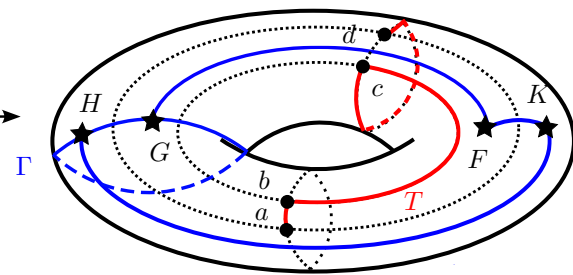
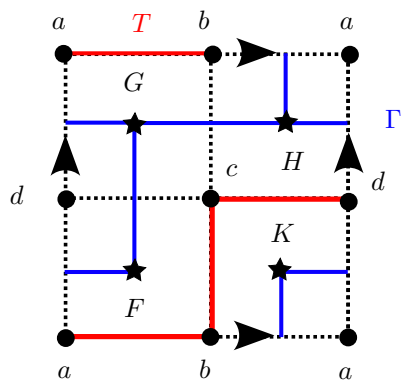
**\*Exercise 5.4.** Now, show that

$$T^2 \# \mathbb{RP}^2 \cong K \# \mathbb{RP}^2 \cong \mathbb{RP}^2 \# \mathbb{RP}^2 \# \mathbb{RP}^2.$$

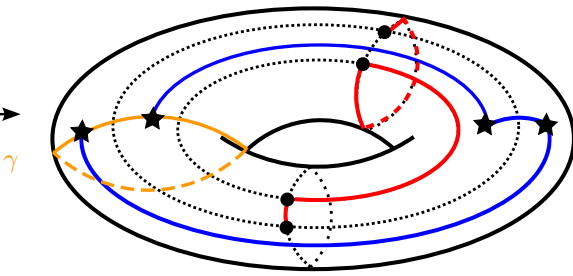
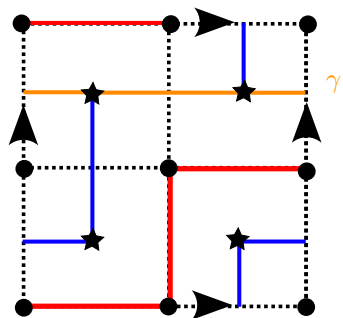
There's a neat video illustrating this on YouTube (<https://youtu.be/aBbDvKq4JqE>).

**\*\*Exercise 5.5.** Generalize the argument in the previous section to prove Theorem 5.2. I haven't put much thought into this, so it may be anywhere from super easy to impossible. I will bring in cookies or something if one of you succeed in doing this one.

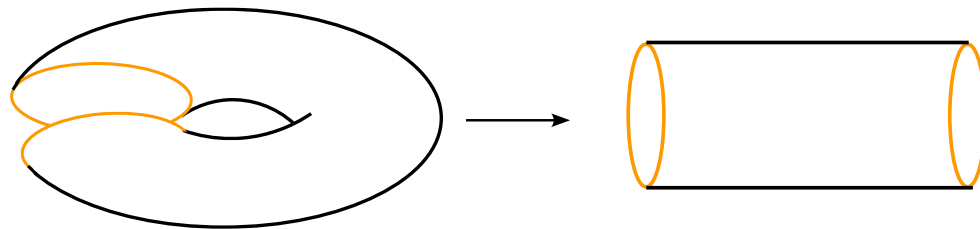
A more complicated cell decomposition of the torus with 4 vertices  $a, b, c, d$ , 8 edges and 4 faces  $F, G, H, K$



Identify one possible loop  $\gamma$  in  $\Gamma$



Cut along  $\gamma$  to get a surface with 2 circle boundary components



“Cap off” the circles with disks to get a surface with a strictly lower Euler characteristic

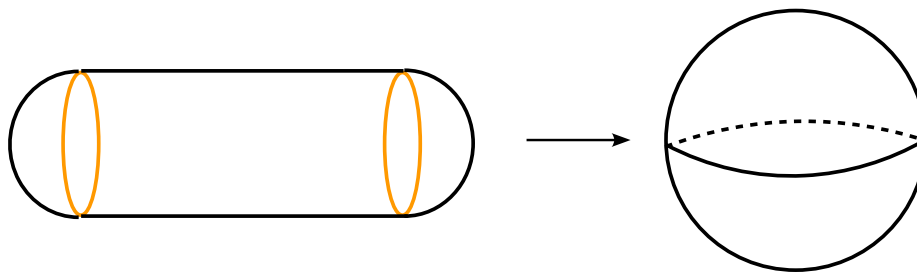


Figure 5.4: An illustration of the process of identifying a loop, cutting, and gluing to cap off

## 6 Riemann Surfaces

### 6.1 Complex Structures

### 6.2 Uniformization

## 7 Basic Algebraic Topology

I've received some requests to do a brief jaunt through the basics of a field closely related to low-dimensional topology, called *algebraic topology*. Whenever you see the names of two subjects in mathematics squished together like this, you should think of the sum of those words as the study of the latter using tools from the former. In this case, we will attempt to study *topology*, as we've been doing, using tools from *abstract algebra*. As the name would imply, abstract algebra is the study of *algebraic structures*, for example groups, rings, modules, vector spaces (as in *linear algebra*), and algebras. These are the objects one would study in our courses Algebraic Structures I and II. The power of algebra comes from the adjective *abstract*. The basic idea is to vastly generalize things we might be interested in studying (numbers, polynomials, symmetries, functions, etc.) by noticing that many of these things share common algebraic properties. We define these abstract algebraic structures, prove abstract statements about these structures, and prove that things we care about are concrete examples of the abstract definitions and apply our theorems in earnest.

Keeping with the theme of these notes, I will try to introduce these methods without getting too into the technical nitty gritty of algebra — seeing as it is a vast, prolific area of study and active research. Although some of you may have seen groups before, I will not assume any familiarity in these notes.

### 7.1 Fundamental Groups

We begin with an incredibly useful *invariant* of manifolds (and topological spaces more generally) called the *fundamental group*. This is an algebraic invariant — a group — which records data about the basic shape of a manifold, for example the number of “holes” it has (in the 2D case this is literally the genus). As with concepts we've worked with previously, we will be concerned with *loops* in a manifold. We will need to modify the concept a bit to really nail down the algebraic structure.

From here on, let  $X$  be an  $n$ -manifold.

**Definition 7.1.** If  $p$  is a point in  $x$ , a *loop based at  $p$  in  $X$*  is a loop  $\gamma : [0, 1] \rightarrow X$  such that  $\gamma(0) = \gamma(1) = p$ . In other words,  $\gamma$  is a loop which starts at  $p$  and ends at  $p$ .

Remember, when we're dealing with loops, we're not always dealing with the nicest case. The loop  $\gamma(t) = p$  for all  $t \in [0, 1]$  is a completely valid loop based at  $p$ . The image of  $\gamma$  might do tons of awful things, including intersect itself however many times it wants.

**Definition 7.2.** A *homotopy* is a continuous deformation between two paths. More formally, if  $\gamma, \gamma'$  are paths starting at  $p$  and ending at  $q$  in  $X$ , then one can think of a homotopy as a smooth family of maps  $\Gamma_t : [0, 1] \rightarrow X$  such that  $\Gamma_0 = \gamma$  and  $\Gamma_1 = \gamma'$ . That is, a family of loops starting at  $\gamma$  and ending at  $\gamma'$  after time has elapsed. If there exists a homotopy from  $\gamma$  to  $\gamma'$ , we say that  $\gamma$  is *homotopic* to  $\gamma'$ , or  $\gamma \simeq \gamma'$  for short.

This is exactly like what we saw in the definition of orientability, except we don't require  $\Gamma_t$  to be a diffeomorphism for any  $t$ .

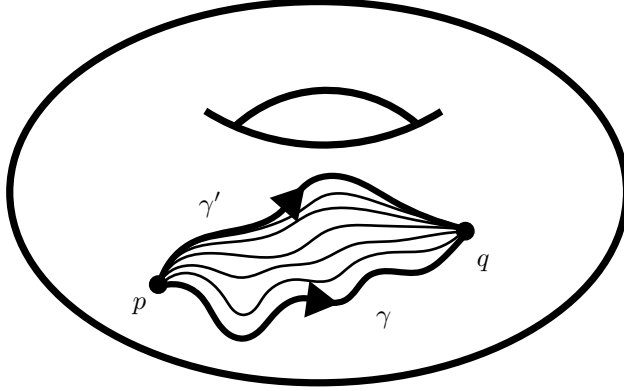


Figure 7.1: Paths  $\gamma, \gamma'$  in the torus and a homotopy  $\Gamma_t$  between them. Notice the arrows indicating the direction a point might flow as time ranges from 0 to 1.

Just as we consider manifolds *up to* diffeomorphism, we will consider loops *up to* homotopy. Remember, objects in topology which can be “continuously deformed” into one another (with the correct notion of “deformed”) are considered the same.

**Definition 7.3.** We define the set  $\pi_1(X, p)$  (which will become our fundamental group) to be the set of all loops in  $X$  based at  $p$  up to homotopy. That is, if two loops  $\gamma$  and  $\gamma'$  are homotopic, then they represent the same element of  $\pi_1(X, p)$ .

If we have two loops based at the same point, it turns out that we have a natural way to “compose” them into a new loop.

**Definition 7.4.** Let  $\gamma, \gamma'$  be loops in  $X$  based at  $p$ . We can define their *product loop*  $\gamma \cdot \gamma' : [0, 1] \rightarrow X$  as

$$(\gamma \cdot \gamma')(t) = \begin{cases} \gamma(2t) & 0 \leq t \leq \frac{1}{2} \\ \gamma'(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}.$$

In words, we start at  $p$ , walk around  $\gamma$  twice as fast as we normally would ending back at  $p$ , then walk around  $\gamma'$  twice as fast to once again end at  $p$ .

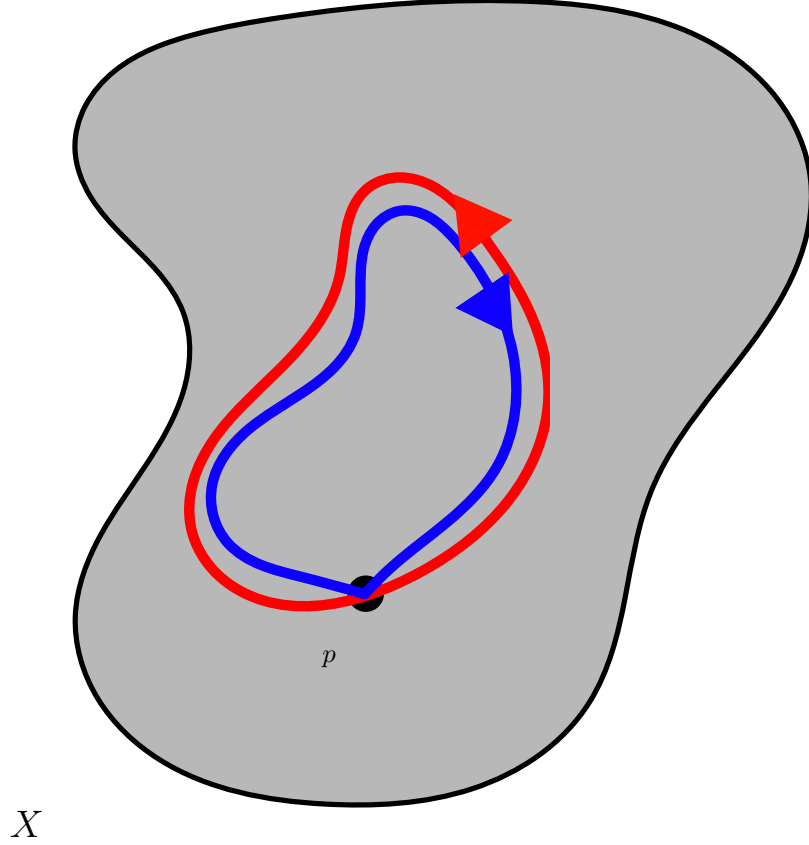


Figure 7.2: The product of loop  $\gamma$  and  $\gamma'$

**Theorem 7.1.** For any point  $p \in X$ , the set  $\pi_1(X, p)$  with the product  $\cdot$  is a group. That is, it satisfies the general requirements of a group:

- (1) *Identity:* For any loop  $\gamma$  based at  $p$ , composing it with the *constant loop*  $c_p$  (which stays at  $p$  for its entire duration) gives a loop which is homotopic to  $\gamma$ . You can think of  $c_p$  as the number 0 in the real numbers with respect to addition, or 1 with respect to multiplication. An important technical note. Although the image of  $\gamma \cdot c_p$  is exactly the image of  $\gamma$  in  $X$ , it is not *literally* the same loop as  $\gamma$ . If we were ants living in  $X$  and following the loop  $c_p \cdot \gamma$ , we would stand still for half the time then rush around  $\gamma$  at twice the normal speed. Similarly, if we followed the loop  $\gamma \cdot c_p$ , we would rush around  $\gamma$  then sit at  $p$  for half the time. The important thing is that these loops are *homotopic*, hence represent the same element of  $\pi_1(X, p)$ .
- (2) *Inverses:* For any loop  $\gamma$  based at  $p$ , there is an “inverse” loop  $\gamma^{-1}$  based at  $p$  such that  $\gamma \cdot \gamma^{-1} \simeq \gamma^{-1} \cdot \gamma$  is homotopic to  $c_p$ . In particular, we can define  $\gamma^{-1}$  to be the loop  $\gamma$  traversed in the opposite direction (flipping the arrow in the above diagrams of loops). Formally,

$$\gamma^{-1}(t) := \gamma(1 - t).$$

- (3) *Associativity:* If  $\gamma_0, \gamma_1, \gamma_2$  are loops in  $X$  based at  $p$ , then

$$(\gamma_0 \cdot \gamma_1) \cdot \gamma_2 \simeq \gamma_0 \cdot (\gamma_1 \cdot \gamma_2).$$

That is, traversing  $\gamma_0$  and  $\gamma_1$  at quadruple speed then  $\gamma_2$  at double speed is the same as traversing  $\gamma_0$

at half speed and then around  $\gamma_1$  and  $\gamma_2$  at quadruple speed.

*Proof Sketch.* I will give the ideas for the formal proof of all of these properties.

- (1) On first reading, you may wonder what there is to prove here. The key observation is the one stated in the technical note above,  $\gamma \cdot c_p$  is not *literally*  $\gamma$ . But it is *homotopic* to  $\gamma$ , and that is what one must prove. The explicit homotopy is not complicated, but it takes some practice if you don't have intuition about these "small" loops which are parametrized in weird ways.

**Exercise 7.1.** Prove that  $\gamma \cdot c_p$  is homotopic to  $c_p \cdot \gamma$  is homotopic to  $\gamma$ .

- (2) This one is tricky to visualize. Instead of working with symbols, I will attempt to draw a picture of the homotopy from  $\gamma \cdot \gamma^{-1}$  to  $c_p$  by representing  $\gamma^{-1}$  slightly perturbed away from where it actually should be in the picture.

□

## 7.2 Seifert-Van Kampen

## 7.3 Cellular Homology

## 8 Knot Theory

Now, I am sure many of you reading this expected me to say, “now that we’re done with dimensions 0, 1, and 2, we’re going to move on to dimension 3!”. Technically, this is true. However, we’re going to have to take a quick excursion back into dimension 1 before we get there. We will briefly explore the subject of *knot theory*. We have seen that dimension 1 is very boring topologically, so you may wonder, why is there a whole theory dedicated to 1 dimensional topological objects?

The difference between classifying 1-manifolds and classifying knots is our notion of *equivalence*. We consider manifolds up to diffeomorphism, but we consider knots up to a different type of topological equivalence, *isotopy*. We’ve actually already seen isotopy! When I said “a smooth family of diffeomorphisms of the disk” to define orientation preserving- and reversing-curves, I meant an isotopy of the disk through the surface. Before I give any formal definitions, let’s look at some examples to gain some intuition: The definition of a knot is relatively straightforward.

**Definition 8.1.** A *knot* is a smooth map  $\gamma : S^1 \rightarrow \mathbb{R}^3$  or, sometimes, a map  $\gamma : S^1 \rightarrow S^3$  where  $S^3$  is the 3-sphere (we will think about this more later), such that  $\gamma$  is a diffeomorphism onto its image. Practically, we will think about knots as the image of these maps, or submanifolds of a 3-dimensional space which are diffeomorphic to  $S^1$ . A **link** is a union of disjoint but possibly interlinked knots.

In the colloquial sense, most people would probably think of knots as tied into a piece of string, like a shoelace. However, these are mathematically pretty boring, since we can always just push either end of the shoelace along the knot to undo it into an unknotted closed interval. So, we consider the case where we *glue* either end of the shoelace together, forming a closed loop which cannot be so easily undone. With this notion of “knot”, how do we say formally when two knots or links are equivalent?

**Definition 8.2.** Two knots  $K$  and  $J$  are equivalent as knots if one can smoothly distort one knot into the other without breaking it. Such a distortion is an example of *ambient isotopy*. More formally, let  $g$  and  $h$  be smooth maps into a manifold  $M$ . An **ambient isotopy** is a smooth family of diffeomorphisms<sup>16</sup>  $F_t : M \rightarrow M$  with  $0 \leq t \leq 1$ , such that  $F_0$  is the identity on  $M$ ,  $F_t$  is a diffeomorphism for all  $t$ , and  $F_1 \circ g = h$ .

An example you can think back to which is specifically an *ambient* isotopy is in our proof that connected sum is well defined. In the proof, we composed three diffeomorphisms to prove that the connected sum is the same regardless of the choice of disks we remove and glue along. The second of these diffeomorphisms secretly came from an ambient isotopy! We smoothly deformed the whole surface in order to slide the first disk to the second, and this is what you should have in mind when someone says *ambient*. But why do we care about how a knot moves around  $\mathbb{R}^3$  when we deform the knot? As it turns out, if we didn’t care what happened to the space around a knot, all knots would be trivial! That is, all knots are isotopic (not ambiently) to the unknot. 8-isotopy-to-unknot It is relatively difficult to tell if two knots are the same or not. 8-gordian-knot The two knots in the figure above are actually the same! The *Gordian knot* is just a picture of the unknot in a really messed up jumble. We need a way to get a handle on knots and to systematically evaluate when knots are the same.

### 8.1 Knot Diagrams and Reidemeister Moves

All of the pictures of knots that we’ve seen so far have been disingenuous. You probably intuitively knew what I meant when I drew something like this: 8-trefoil-knot But if you really think about it, what does the

<sup>16</sup>If you’re curious about what this means formally, a “smooth family of diffeomorphisms” is actually just a smooth map  $F : [0, 1] \times M \rightarrow M$



picture mean? Knots live in 3-dimensional space, and this picture is clearly a representation of the knot in a 2D plane (the page it's drawn on). Generally, this is how mathematicians think of knots — as their 2D projections.

**Definition 8.3.** Given a knot  $K$  in  $\mathbb{R}^3$ , its **projection** is its image under a map like  $(x, y, z) \mapsto (x, y)$ , smoothly mapping all of  $\mathbb{R}^3$  onto  $\mathbb{R}^2$ . We also include some extra data: at each intersection of  $K$  with itself (called **crossings**) under the projection, we remember which strand falls on top of the other. We denote this using one of the following pictures at the intersection: 8-local-view-of-intersection We call these pictures **knot diagrams**.

Technically, in order for this definition to make sense, we should add an extra condition that the “intersections” of the knot with itself under a projection are *transverse*,<sup>17</sup> or that there are only finitely many of them. Generally speaking, we can always do this with some kind of small ambient isotopy pushing the two overlapping strands over each other to achieve *transversality*.

Using these pictures, we can turn all questions about knots in 3-space up to ambient isotopy into combinatorial ones about knot diagrams in the plane. But up to what notion of equivalence do we consider knot diagrams?

**Definition 8.4.** Two knot diagrams are the same if we can change one into the other using *planar isotopy* and *Reidemeister moves*. **Planar isotopy** is simply an ambient isotopy of the diagram in  $\mathbb{R}^2$ , where crossings are treated as a single point. It is fairly clear why this represents an ambient isotopy back in  $\mathbb{R}^3$  — just do the same isotopy while keeping in mind perspective. There are three other ways to manipulate knot diagrams, and they all tell us how we can change up its crossings. **Reidemeister moves** are the three ways we can change the crossings of a knot while keeping the knot the same. These moves look like 8-reidemeister-moves Again, it is pretty straightforward how each of these can be interpreted as ambient isotopy in  $\mathbb{R}^3$ .

Less trivial though is the fact that:

**Theorem 8.1.** All ambient isotopies of the knot can be represented using planar isotopy and Reidemeister moves in an associated knot diagram.

A detailed proof of this theorem can be found in e.g. *Knot Theory and Its Applications* by Kunio Murasugi.<sup>18</sup> This gives us a systematic way of checking whether two diagrams represent the same knot, since we can go between any knot diagrams representing the same knot using these moves.

## 8.2 Tricolorability and Knot Invariants

Unfortunately, finding all of the isotopies and Reidemeister moves to turn one knot diagram into another is *also* quite difficult.<sup>19</sup> So, we employ the same strategy we did with manifolds: we use *invariants*. There are many, *many* knot invariants out there, each more complicated than the last. In this section, we'll explore a relatively simple one, *tricolorability*.

**Definition 8.5.** A **strand** in a knot diagram is a piece of the knot that goes from one under crossing to another. A knot is **tricolorable** if each strand of the knot diagram can be colored one of three colors, subject to the following rules:

- (1) At least two colors must be used, and

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<sup>17</sup>We'll talk much more about this later.

<sup>18</sup><https://www.maths.ed.ac.uk/~v1ranick/papers/murasugi3.pdf>

<sup>19</sup>Actually, it is a major unsolved problem as to how hard this actually is. Even determining if a knot diagram is for the unknot is not known to admit a polynomial time algorithm, and it's possible that this problem is *NP-complete*.

(2) At each crossing, the three incident strands are either all the same color, or all different colors.

For example, the trefoil knot and trivial 2-link are tricolorable, but the unknot and figure-eight knot are not.

**Theorem 8.2.** Tricolorability is a knot diagram invariant — either every projection of a knot is tricolorable, or none of them are. This is equivalent to saying that applying planar isotopy and/or Reidemeister moves to a knot diagram preserves tricolorability.

**Exercise 8.1.** Prove Theorem 8.2.

These combinatorial diagrams and strategies for defining invariants according to “moves” is ubiquitous in low-dimensional topology. More than likely, we will see this strategy a number of times throughout the next few sections. Diagrams will be our main way of getting our hands on manifolds we can’t directly visualize or see, like 3- and 4-manifolds.

## 9 3-Manifolds

This is the first section on objects that most readers probably won't have encountered before in some capacity. These objects are also the first that are objectively very difficult to visualize, given that humans don't have a good 4-dimensional imagination (at least Klein bottles and  $\mathbb{RP}^2$  have *something* we can look at in 3 dimensions).

A 3-manifold is a space which we can think of as locally looking like the physical space we inhabit. If we were small enough, we could sit inside of a 3-manifold and not be able to notice a difference between  $\mathbb{R}^3$  and that space — it's even possible that our universe has some interesting global topology that we're too small to observe! We could be living in a 3-sphere or 3-torus.

As is usually the case in this subject, dimension three has some very exciting specific behavior that doesn't generalize well to other dimensions. There are whole areas of research dedicated specifically to tools and techniques to study and understand 3-manifolds, many of which are connected to other theories like knot theory, geometric group theory, hyperbolic geometry, number theory, topological quantum field theory, gauge theory, Floer homology, PDEs, etc. etc.

Just as we used embedded curves to understand surfaces, it is common in dimensions 3 and 4 to use embedded *surfaces* to understand 3- and 4-manifolds. This will lead to the idea of *incompressible surfaces* and the theory of Haken manifolds, or if one chooses pieces of a 3-manifold to be particularly nice, to structures like *Heegaard splittings*, which admit combinatorial descriptions much like knot diagrams.

### 9.1 Examples of 3-Manifolds

### 9.2 Visualizing $S^3$

### 9.3 The Hopf Fibration

## 10 Handle Calculus in Dimension 3

## 11 Dehn Surgery

### 11.1 Constructing More 3-Manifolds

## 12 Geometrization of 3-Manifolds

### 12.1 The Poincaré Conjecture

## 13 Why is Dimension 4 So Weird?

### 13.1 Exotica

### 13.2 The Poincaré Conjecture (Again)

### 13.3 Dimensions 5 and Up: $h$ -Cobordisms

## 14 Handle Calculus in Dimension 4

### 14.1 Knot Traces



## 15 The Classification of Topological 4-Manifolds

### 15.1 Intersection Forms

### 15.2 Classification